Unit IV

Introduction of IC Engines and Automobile

I C Engine

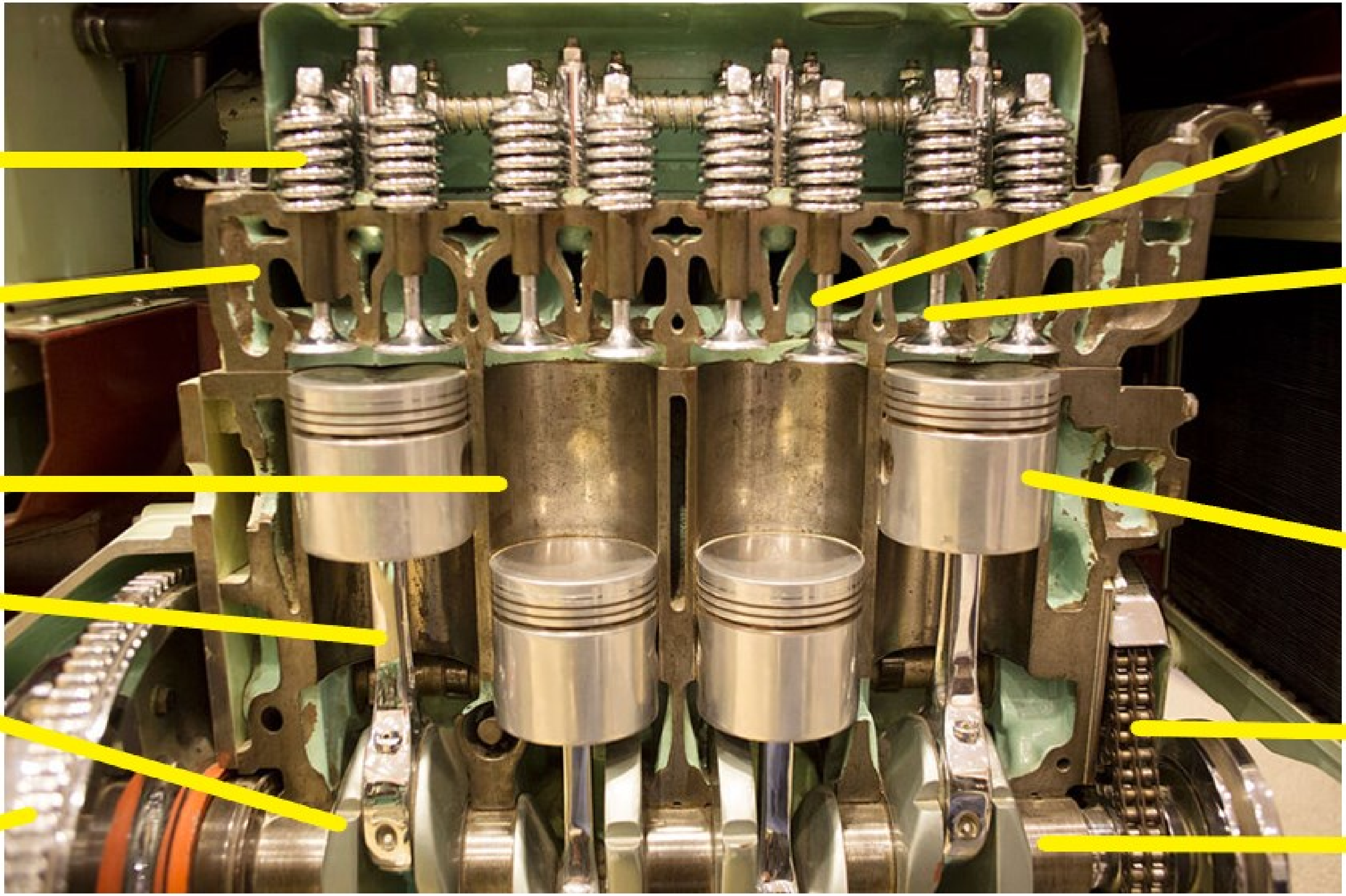
- Internal combustion (IC) engine is a machine which converts heat energy of fuel into mechanical energy.
- The energy can further be utilized for various purposes like for propelling buses, trucks, cars, two wheelers, operating heavy industrial equipments, constructions machineries, ships etc.
- Combustions of fuel takes place inside the engine cylinder.

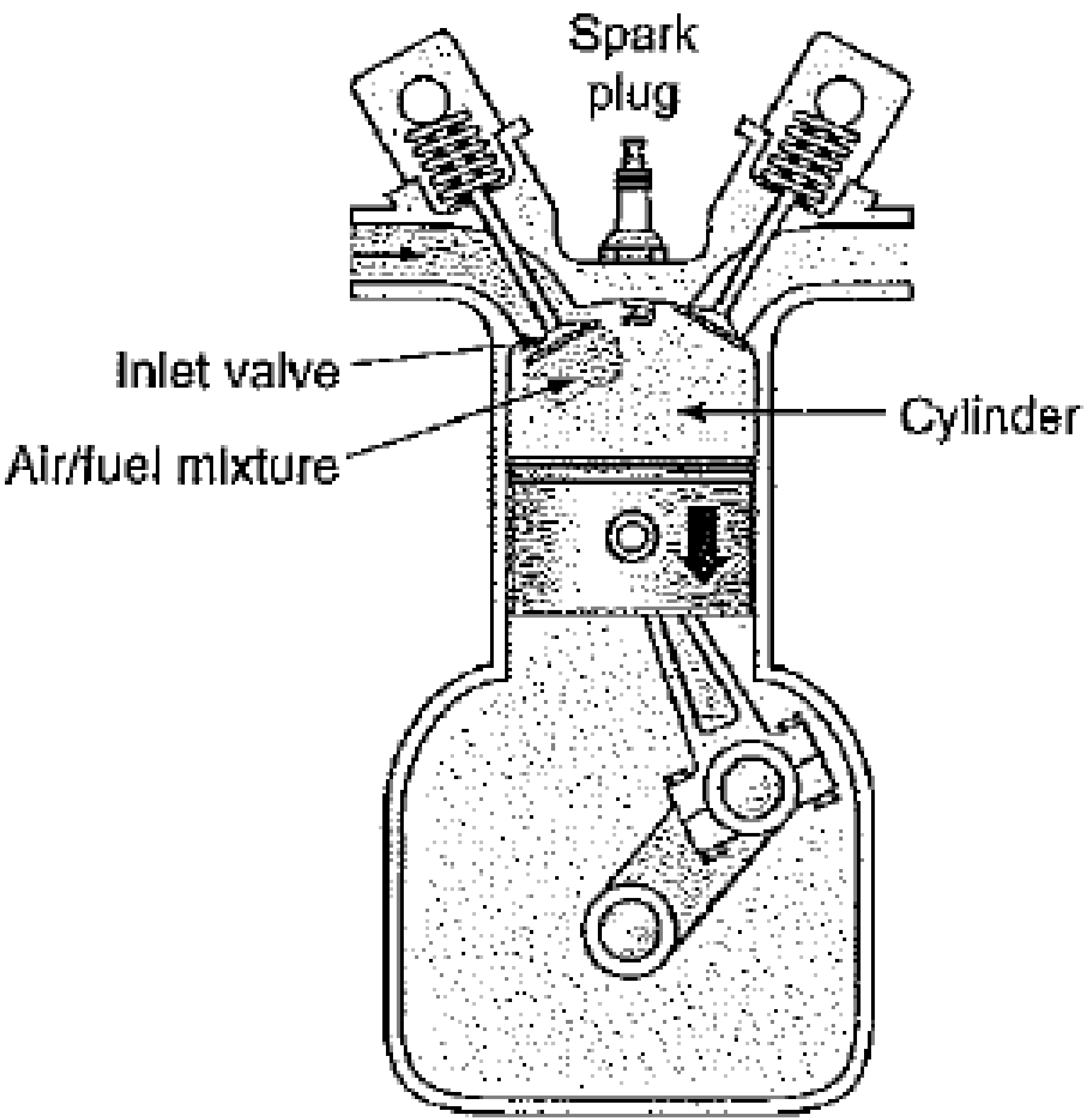
Classification of IC Engine

Different criteria of classification are;

1. On the basis of Fuel used
   i. Petrol Engine (Gasoline engine), ii. Diesel Engine
2. One the basis of Number of Storkes
   i. Two Stroke Engine, ii. Four Stroke Engine
3. On the basis of number of cylinders
   i. Single Cylinder, ii. Multicylinder Engine
4. On the cooling system used
   i. Air cooled, ii Liquid cooled with natural or forced circulation
5. On the basis of orientation of cylinder
   i. Horizontal Cylinder, ii. Vertical Cylinder, iii. Inclined Cylinder
6. Arrangement of Cylinder
   i. Incline Engine, ii. V-Engine, iii. Radial Engine
7. On the basis of Thermodynamic Cycle
   i. Otto cycle, ii Diesel cycle

Main Parts of I C Engine




Spark plug
Inlet valve
Air/fuel mixture
Cylinder

Main Parts of Engine

1. Cylinder: It is a cylindrical block having cylindrical space inside for piston to make reciprocating motion. Upper portion of cylinder which covers it from the top is called cylinder head. This is manufactured by casting process and materials used are cast iron or alloy steel.

2. Piston and Piston rings: Piston is a cylindrical part which reciprocates inside the cylinder and is used for doing work and getting work. Piston has piston rings tightly fitted in groove around piston and provide a tight seal so as to prevent leakage across piston and cylinder wall during piston's reciprocating motion. Pistons are manufactured by casting or forging process. Pistons are made of cast iron, aluminum alloy. Piston rings are made of silicon, cast iron, steel alloy by casting process.

3. Combustion space: It is the space available between the cylinder head and top of piston when piston is at farthest position from crankshaft (TDC).

4. Intake manifold: It is the passage/duct connecting intake system to the inlet valve upon cylinder. Through intake manifold the air/air-fuel mixture goes into cylinder.

5. Exhaust manifold: It is the passage/duct connecting exhaust system to the exhaust valve upon cylinder. Through exhaust manifold burnt gases go out of cylinder.

6. Valves: Engine has both intake and exhaust type of valves which are operated by valve operating mechanism comprising of cam, camshaft, follower, valve rod, rocker arm, valve spring etc. Valves are generally of spring loaded type and made out of special alloy steels by forging process.

7. Spark plug: It is the external ignitor used for initiating combustion process. Spark plug is activated by electrical energy fed by electrical system with engine. It delivers spark with suitable energy to initiate combustion at appropriate time for suitable duration.

8. Bearing: Bearings are required to support crank shaft. Bearings are made of white metal leaded bronze.

9. Connecting rod: It is the member connecting piston and crankshaft. It has generally I section and is made of steel by forging process.

10. Crank: It is the rigid member connecting the crankshaft and connecting rod. Crank is mounted on crankshaft. Crank transfers motion from connecting rod to crankshaft as it is linked to connecting rod through crank pin.

11. Crankshaft: It is the shaft at which useful positive work is available from the piston-cylinder arrangement. Reciprocating motion of piston gets converted into rotary motion of crankshaft. Crankshaft are manufactured by forging process from alloy steel.

12. Crankcase: Crankcase actually acts like a sump housing crank, crankshaft, connecting rod and is attached to cylinder. These are made of aluminium alloy, steel, cast iron etc. by casting process.

13. Gudgeon pin: It is the pin joining small end of the connecting rod and piston. This is made of steel by forging process.

14. Cams and Camshafts: Cams are mounted upon camshaft for opening and closing the valves at right timings and for correct duration. Camshaft gets motion from crankshaft through timing gears.

15. Carburettor : Carburettor is device to prepare the air fuel mixture in right proportion and supply at right time.

Terminology

Bore - Inside diameter of the cylinder
Stroke – Distance travelled by the piston inside the cylinder
RPM – Revolutions per minutes unit of rotational speed
Brake Horse Power – Useful power available at the crank shaft
Indicated Horse Power – Power developed inside the engine cylinder
Mechanical Efficiency – Ratio of brake power to indicated power
Stroke Volume – The volume covered by the piston during a stroke

Dead center: It refers to the extreme end positions inside the cylinder at which piston reverses it’s motion. Thus, there are two dead centers in cylinder, called as ‘top dead center’ or ‘inner dead center’ and ‘bottom dead center’ or ‘outer dead center’. Top dead center (TDC) is the farthest position of piston from crankshaft. It is also called inner dead centre (IDC). Bottom dead center (BDC) refers to the closed position of piston from crankshaft. It is also called outer dead centre (ODC).

Swept volume : It is the volume swept by piston while travelling from one dead center to the other. It may also be called stroke volume or displacement volume. Mathematically,

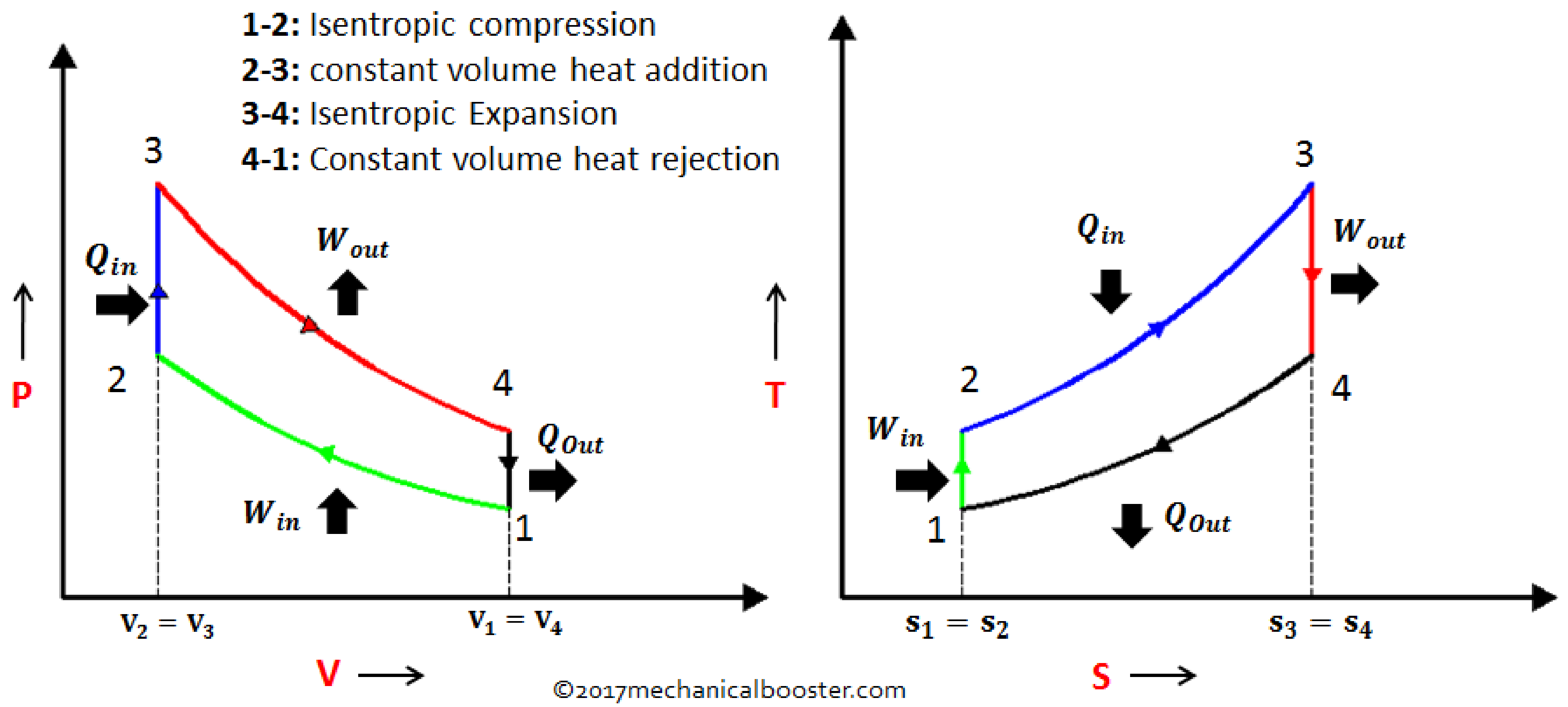
Swept volume = Piston area × Stroke

Clearance volume: It is the volume space above the piston inside cylinder, when piston is at top dead center. It is provided for cushioning considerations and depends, largely upon compression ratio.

Compression ratio: It is the ratio of the total cylinder volume when piston is at BDC to the clearance volume.

Compression ratio = (Swept volume + Clearance volume) / Clearance volume

Otto Cycle



P-V and T-S Diagram of Otto Cycle

Efficiency of Otto Cycle

Thermal Efficiency = $\dfrac{\boldsymbol{Heat\ Supplied - Heat\ Rejected}}{\boldsymbol{Heat\ Suppiled}}$

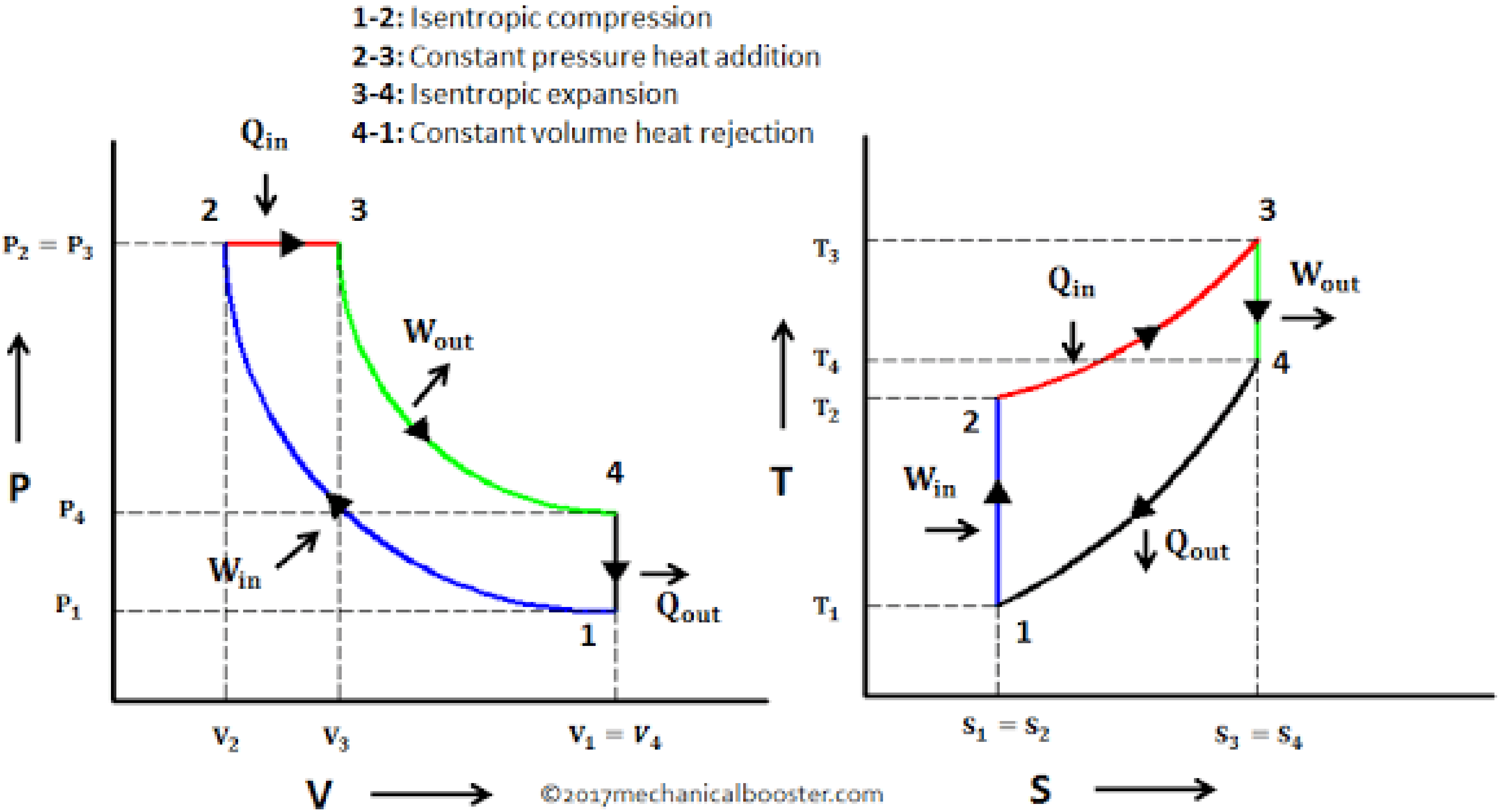
Air Standard Efficiency

$$\eta_{Otto} = 1 - \frac{1}{r^{(\gamma-1)}}$$

r = compression ratio (6 to 10)

γ = Ratio of Specific heat = 1.4

Diesel Cycle



P-V and T-S Diagram of Diesel Cycle

Efficiency of Diesel Cycle

Mechanical Efficiency

$$\eta_{Diesel} = \frac{Q_S - Q_R}{Q_S}$$

Air Standard Efficiency

$$\eta_{Diesel} = 1 - \frac{1}{r^{(\gamma-1)}}\left[\frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)}\right]$$

r = compression ratio (16 to 20)

γ = Ratio of Specific heat =1.4

r_c = Cut off ratio

Power Produced by the Engine

Brake Power BP = $\frac{2\pi NT}{60}$

Indicated Power IP = $\frac{pLAN}{60}$

Where N = Engine RPM
T = Torque available on crank shaft
L = Strole Length
A = Cross section are of piston
p = Mean Effective pressure

An engine working on Otto cycle has the following conditions : Pressure at the beginning of compression is 1 bar and pressure at the end of compression is 11 bar. Calculate the compression ratio and air-standard efficiency of the engine. Assume γ = 1.4.

Solution: For isentropic process $pv^{\gamma} = Constant$

$$r = \frac{V_1}{V_2} = \left(\frac{p_2}{p_1}\right)^{\left(\frac{1}{\gamma}\right)} = 11^{\frac{1}{1.4}} = 5.54$$

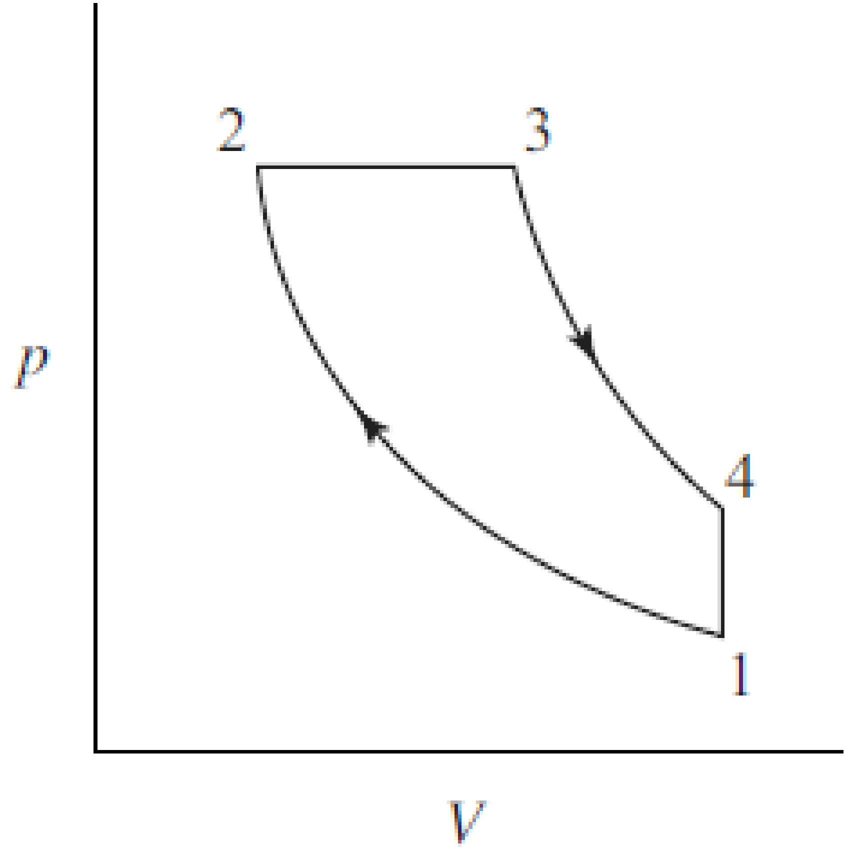
$$\eta_{\text{air-std}} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \left(\frac{1}{5.54}\right)^{0.4}$$

$$= 0.496 = 49.6\%$$

A Diesel engine has a compression ratio of 20 and cut-off takes place at 5% of the stroke. Find the air-standard efficiency. Assume γ = 1.4.

Solution: Cut off ratio = $\dfrac{\text{Volume at the end of injection}}{\text{Volume at the beginning of injection}}$

$$
\begin{aligned}
r &= \frac{V_1}{V_2} = 20 \\
V_1 &= 20V_2 \\
V_s &= 20V_2 - V_2 = 19V_2 \\
V_3 &= 0.05V_s + V_2 \\
&= 0.05 \times 19V_2 + V_2 \\
&= 1.95V_2
\end{aligned}
$$



$$
\begin{aligned}
r_c &= \frac{V_3}{V_2} = \frac{1.95V_2}{V_2} = 1.95 \\
\eta &= 1 - \frac{1}{r^{\gamma-1}} \; \frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)} \\
&= 1 - \frac{1}{20^{0.4}} \times \left[\frac{1.95^{1.4} - 1}{1.4 \times (1.95 - 1)}\right] = 0.649 = \mathbf{64.9}\%
\end{aligned}
$$

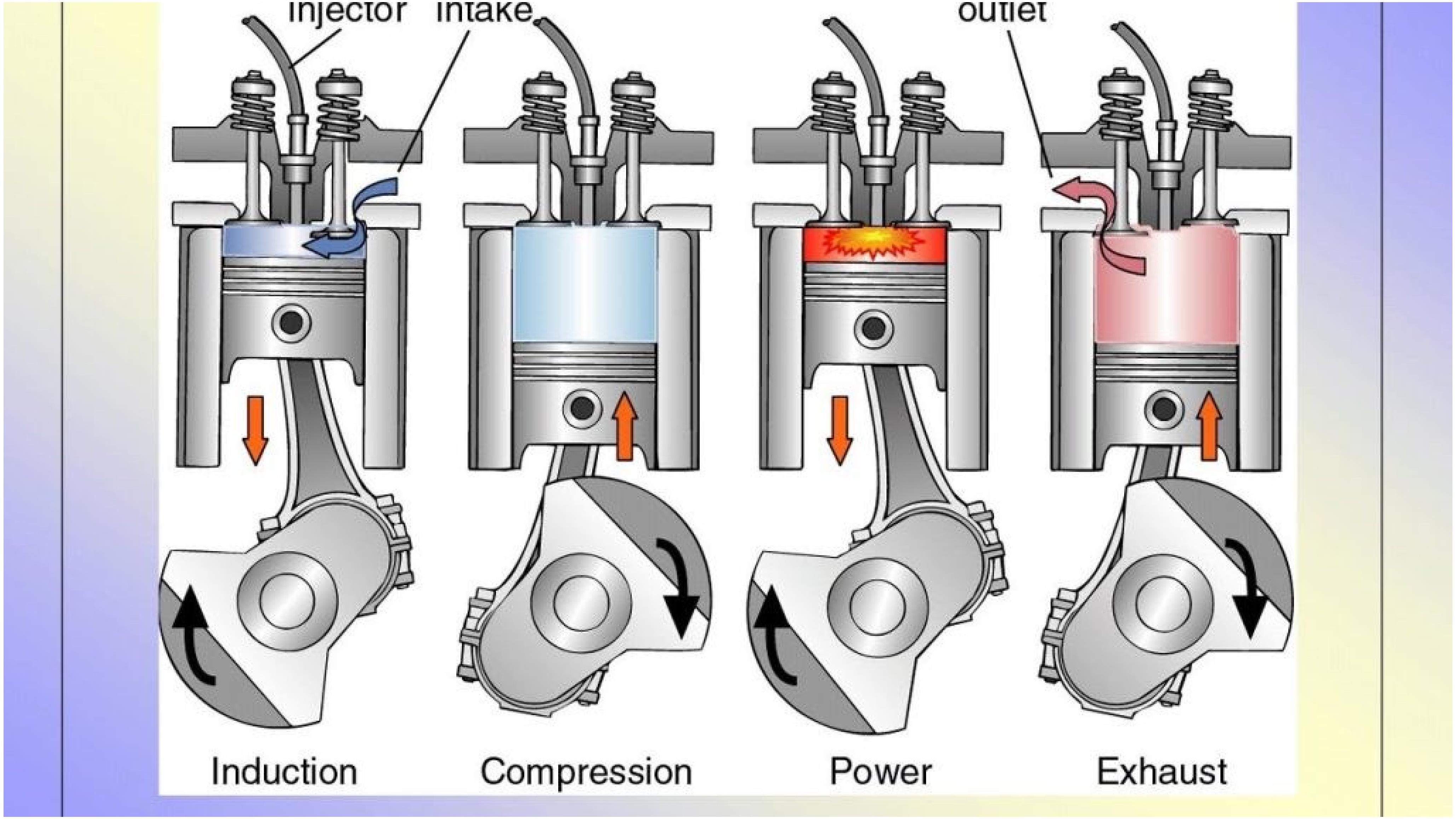
The cubic capacity of a four-stroke over-square spark-ignition engine is 245 cc. The over-square ratio is 1.1. The clearance volume is 27.2 cc. Calculate the bore, stroke and compression ratio of the engine.

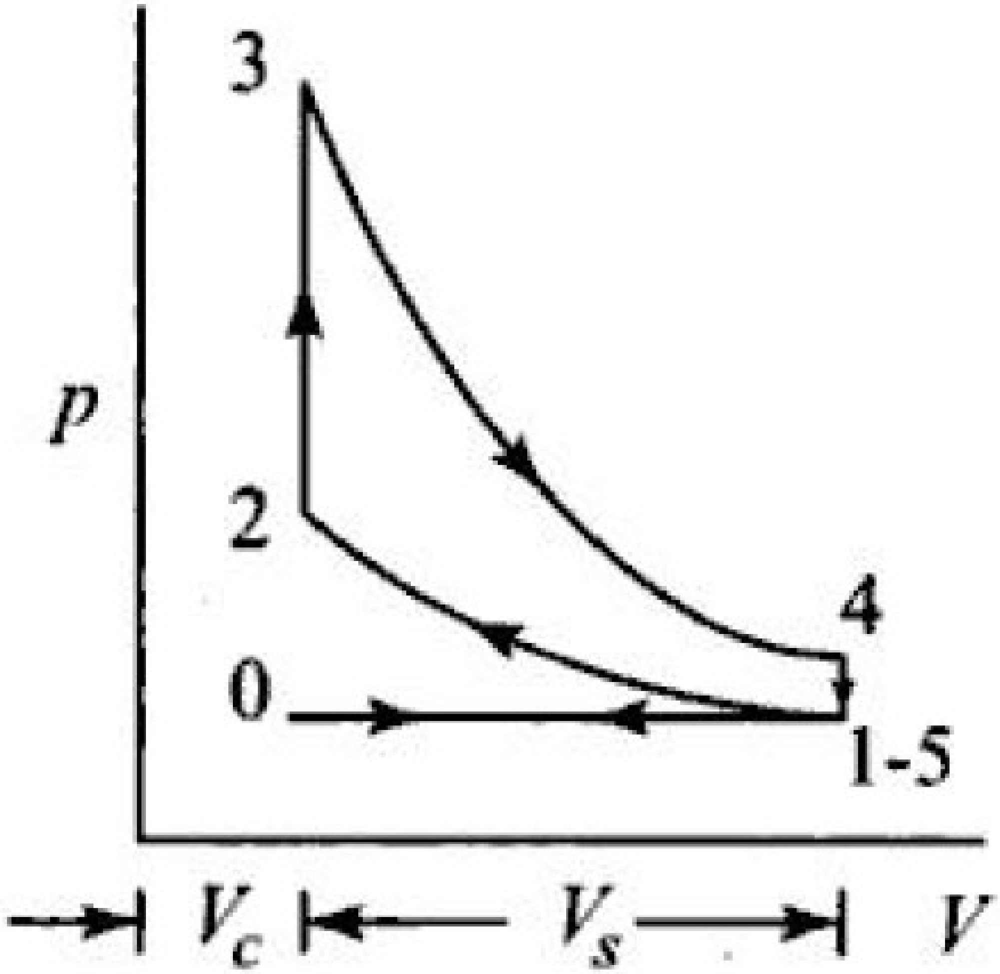
$$
\begin{aligned}
\text{Cubic capacity, } V_s &= \frac{\pi}{4} d^2 L = \frac{\pi}{4}\,\frac{d^3}{1.1} = 245 \\
d^3 &= 343 \\
\text{Bore, } d &= \mathbf{7\ cm} \\
\text{Stroke, } L &= \frac{7}{1.1} = \mathbf{6.36\ cm} \\
\text{Compression ratio, } r &= \frac{V_s + V_c}{V_c} \\
&= \frac{245 + 27.2}{27.2} = \mathbf{10}
\end{aligned}
$$

Processes in the Engine Working Cycle

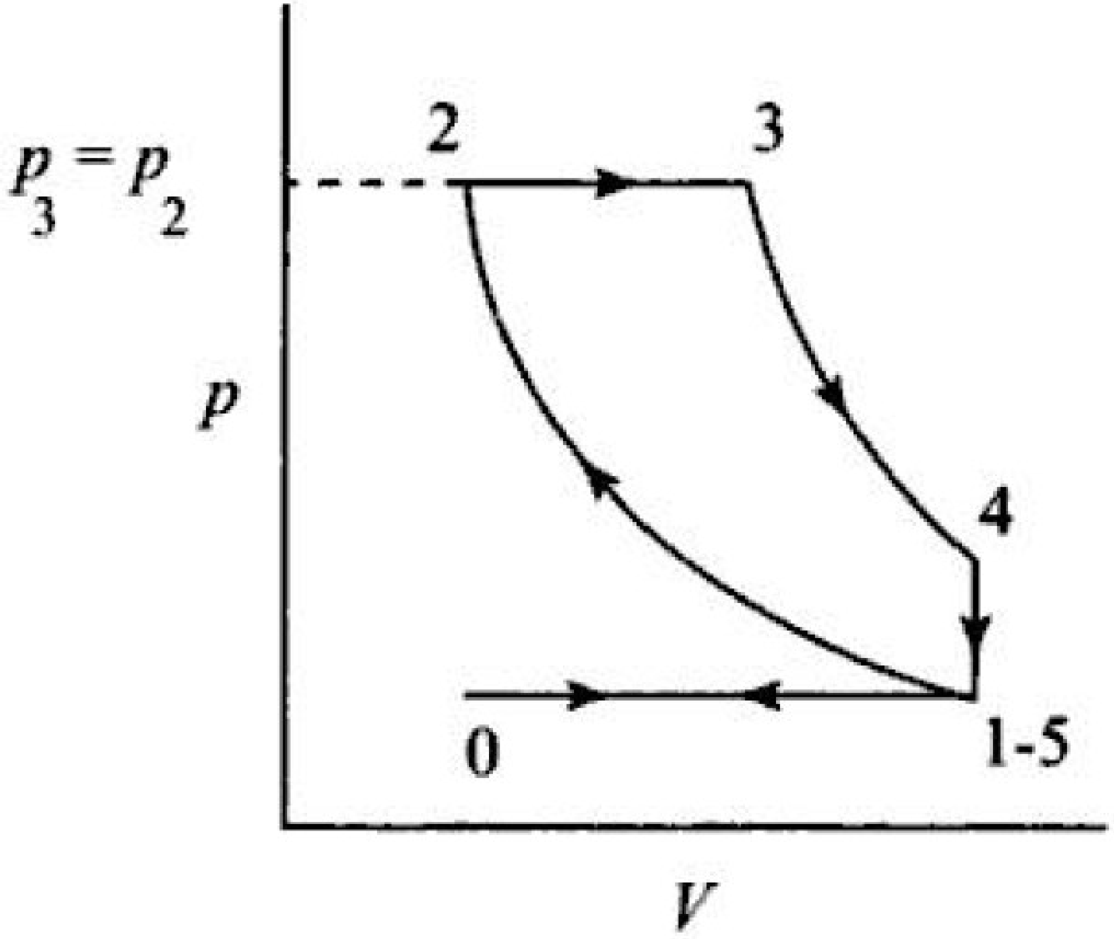
1. **Suction** – In this process air fuel mixture in case of petrol engine or fresh air in case of diesel enters into the engine cylinder through inlet valve.
2. **Compression** – The freshly entered air fuel mixture or only air gets compressed which reduces volume of air and increases pressure and temperature.
3. **Expansion** – At the end of compression process air fuel mixture ignited (in case of petrol engine) or fuel is injected into the cylinder is ignited (in case of diesel engine) which rises pressure and temperature and piston is pushed back.
4. **Exhaust** – After the completion of expansion brunt gases are released into the atmosphere through exhaust valve.

Working of Four Stroke Cycle Engine





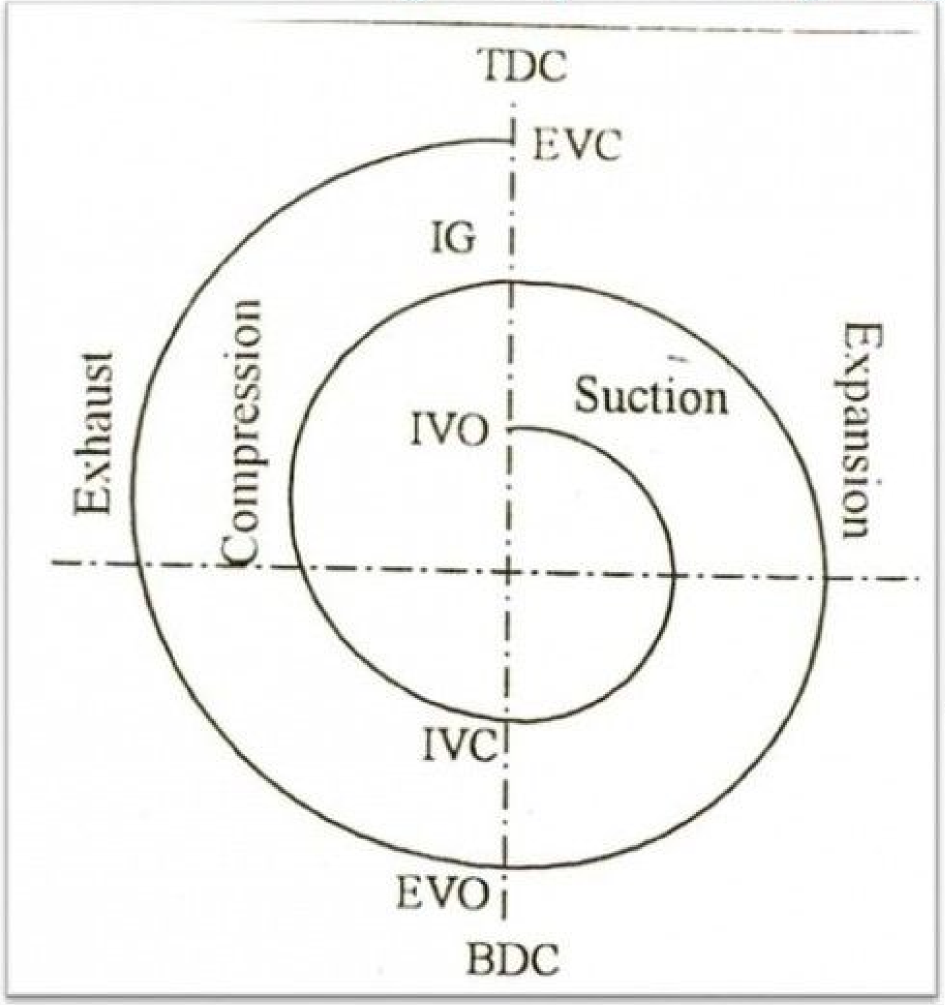
3
p
2
4
0
1-5
V_c
V_s
V



2
3
$p_3 = p_2$
p
4
0
1-5
V

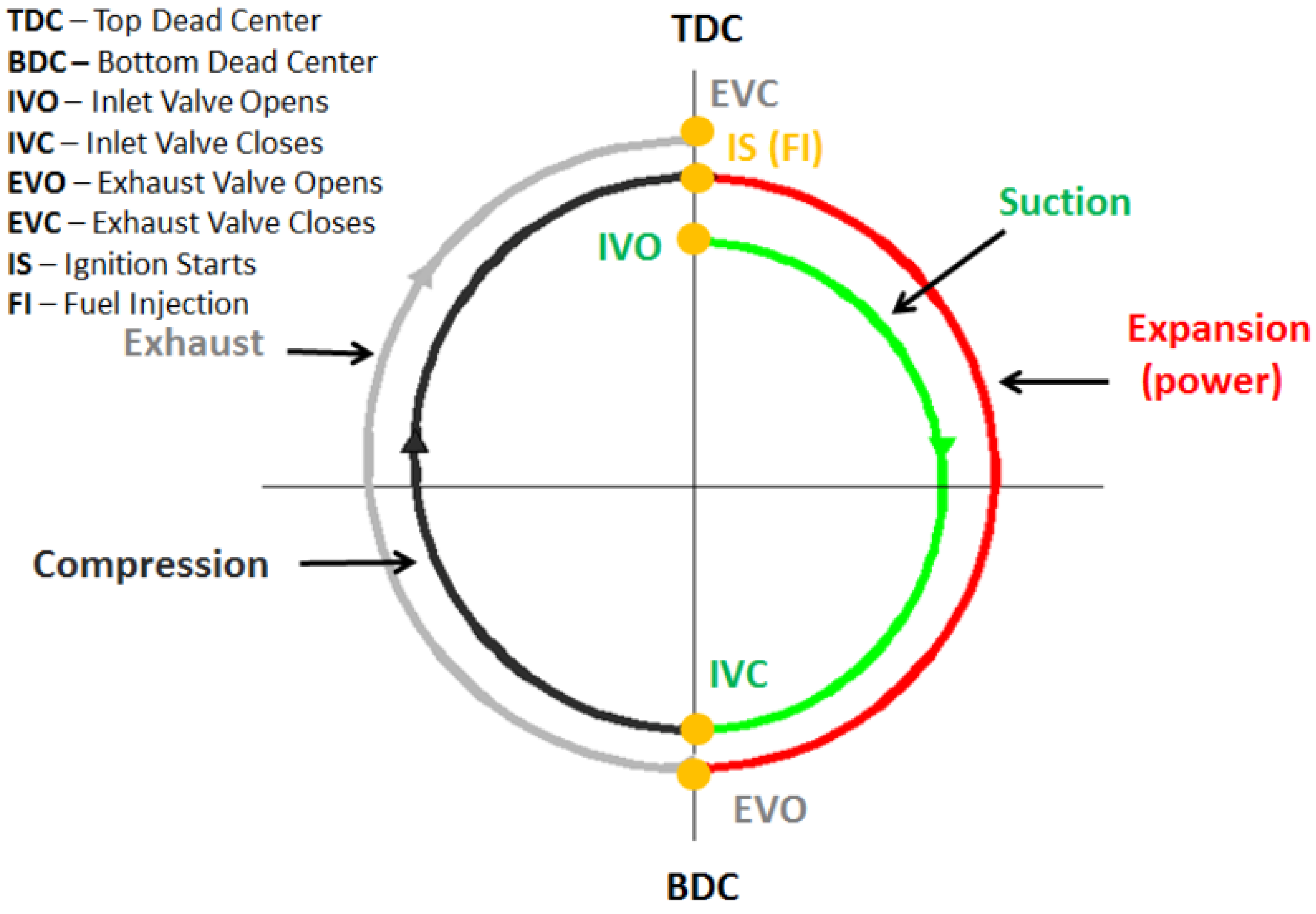
VALVE TIMING AND PORT TIMING DIAGRAMS

❖ Valve timing and port timing diagrams

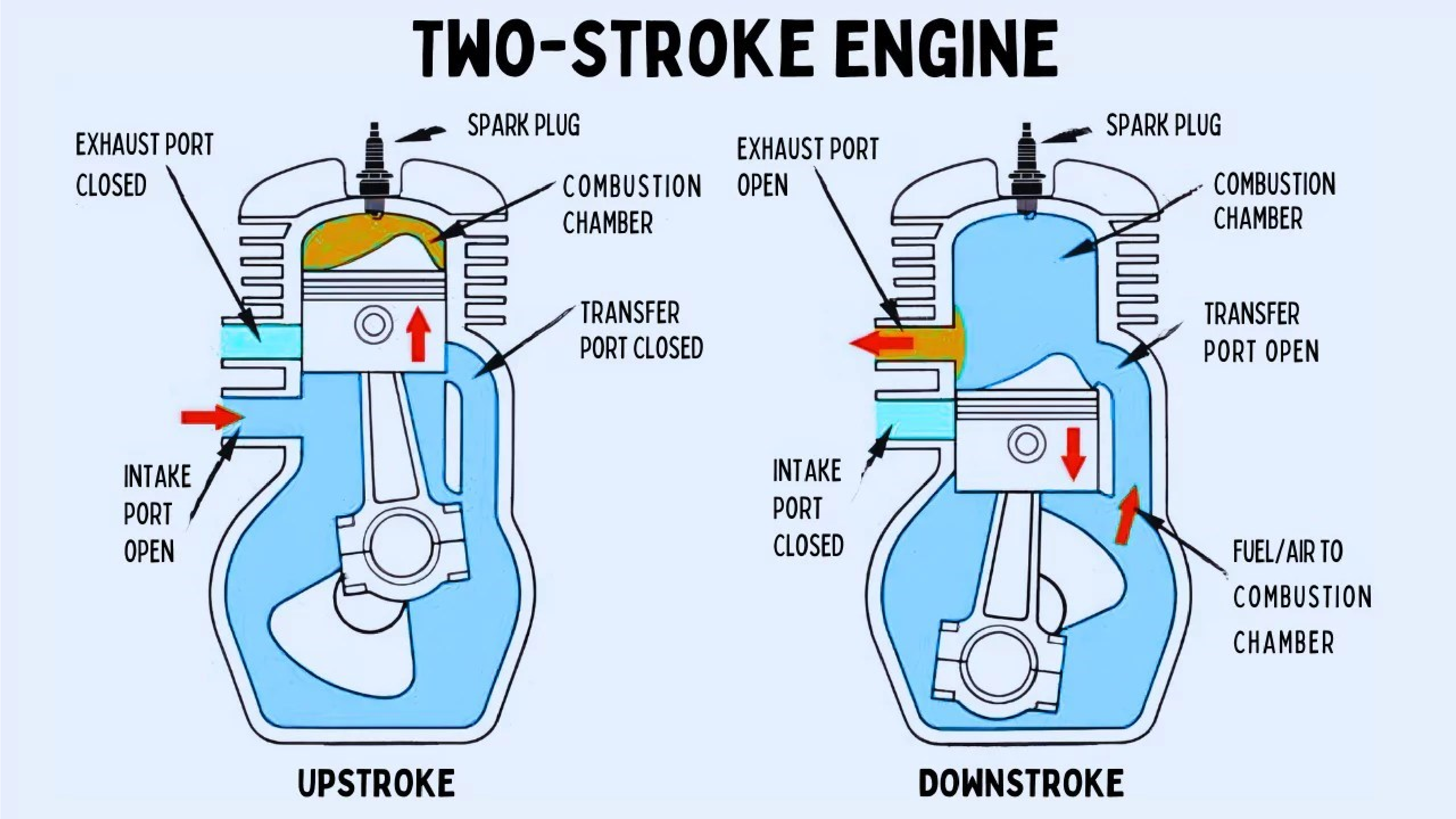


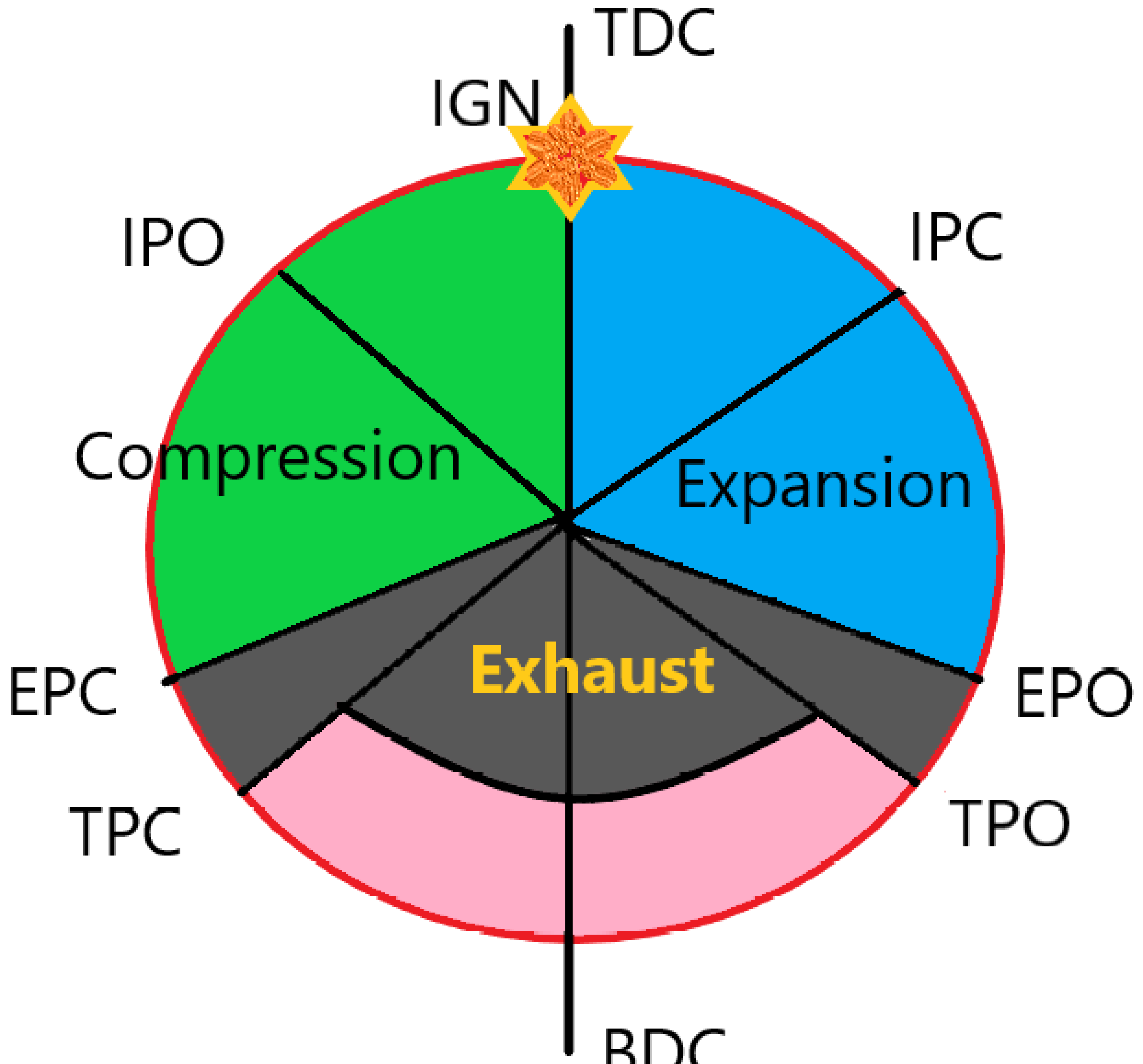
IVO	⇒	Inlet Valve Open.
IVC	⇒	Inlet Valve Close
IS	⇒	Ignition starts
EVO	⇒	Exhaust valve Open
EVC	⇒	Exhaust valve Close
TDC	⇒	Top Dead Center
BDC	⇒	Bottom Dead Center

Theoritical Valve Timing Diagram


TDC – Top Dead Center
BDC – Bottom Dead Center
IVO – Inlet Valve Opens
IVC – Inlet Valve Closes
EVO – Exhaust Valve Opens
EVC – Exhaust Valve Closes
IS – Ignition Starts
FI – Fuel Injection
TDC
EVC
IS (FI)
IVO
Suction
Expansion
(power)
Exhaust
Compression
IVC
EVO
BDC

Working of Two Stroke Engine





IPO – Inlet port Open
IPC – Inlet port Close
EPO – Exhaust port Open
EPC – Exhaust port Close
TPO – Transfer port Open
TPC – Transfer port Close
IGN – Ignition
TDC – Top Dead Center
BDC – Bottom Dead Center

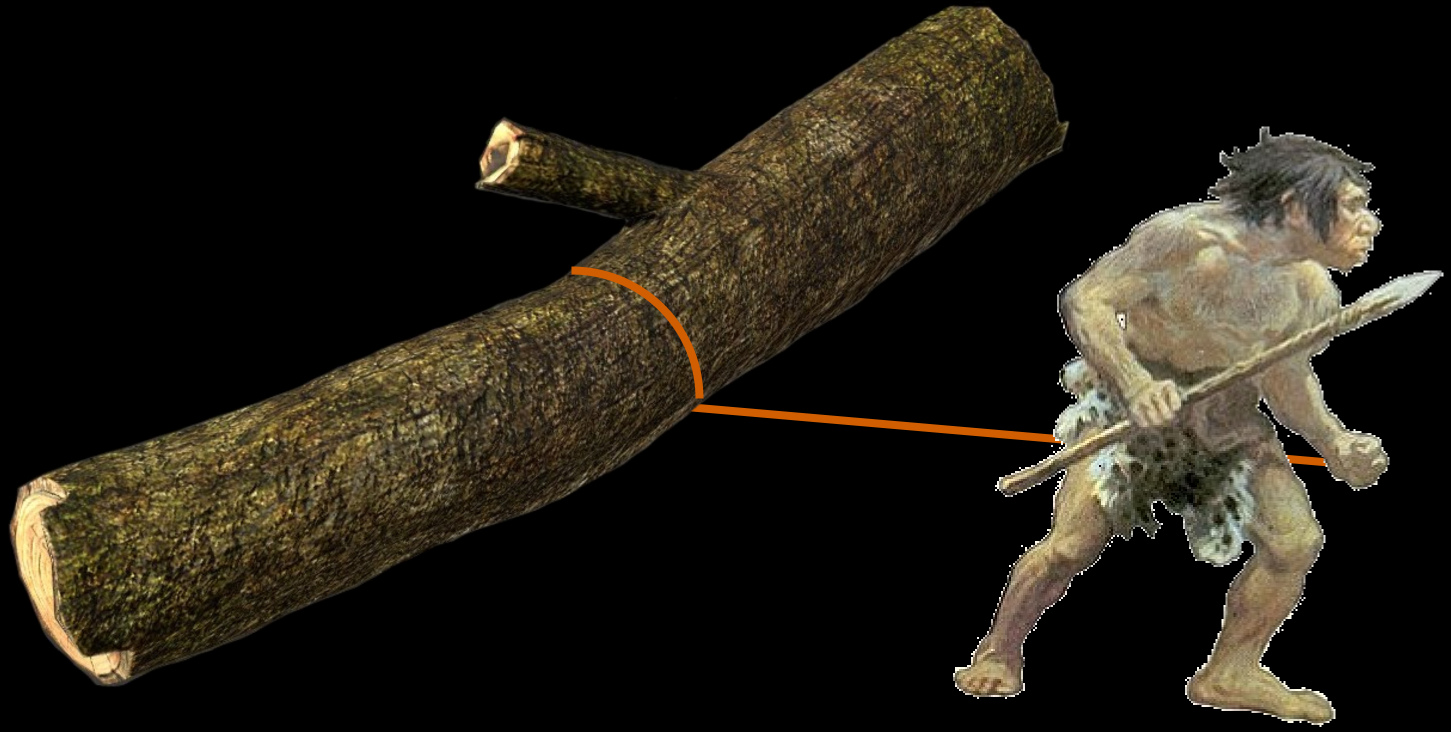
AUTOMOBILE ENGINEERING

Dr. Dinesh Kumar Pasi, Associate Professor, MED, SGSITS, Indore

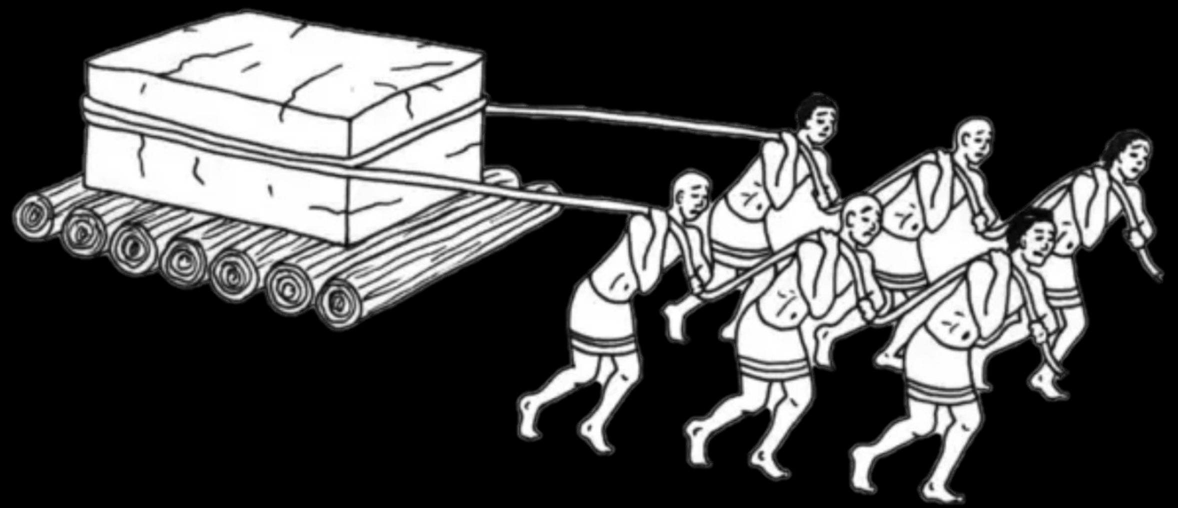
GENERAL OVERVIEW OF VEHICLE

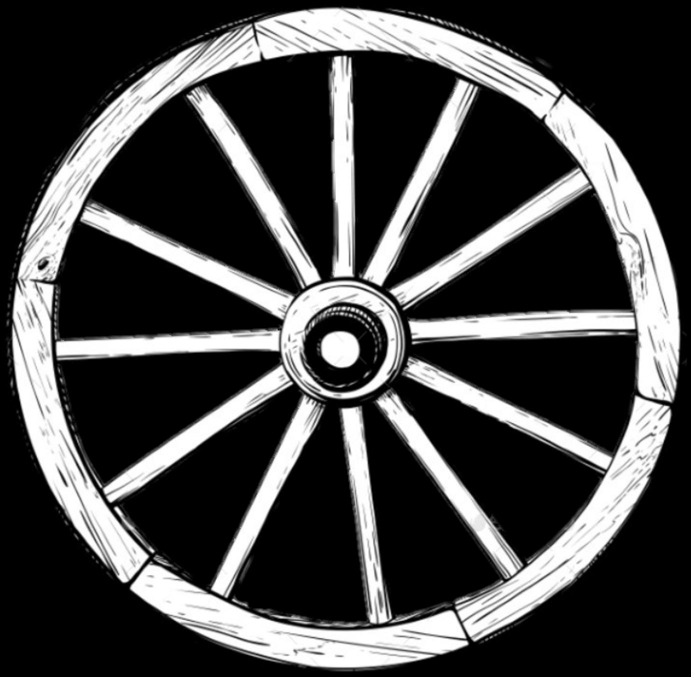
HISTORY OF AUTOMOBILE

Early man used shift heavy loads by dragging instead lifting



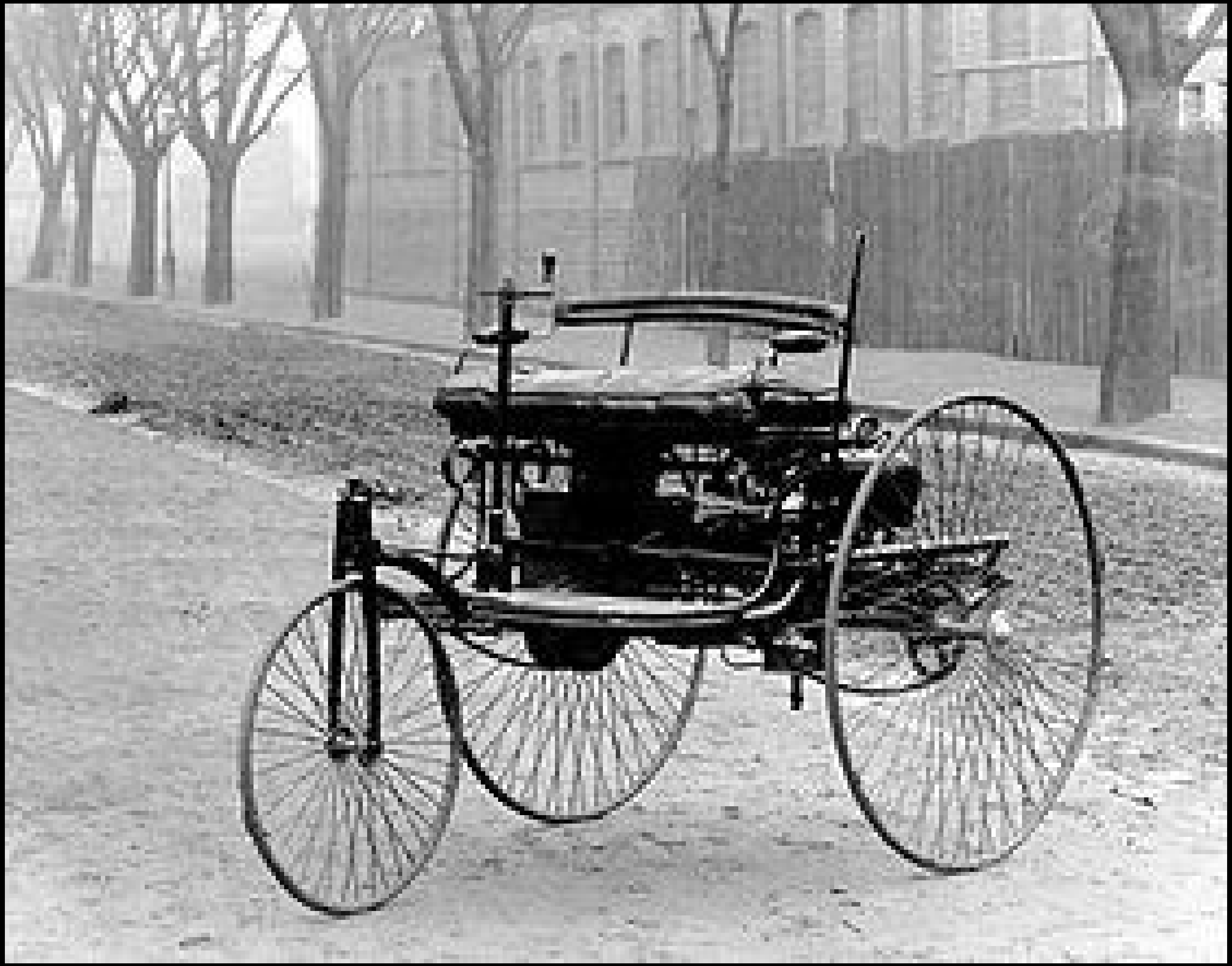
It was further made easy by the use of logs, and it became the forefather of wheel, and then wheel were which modern industry runs and vehicle runs





FIRST EVER CAR

In the year 1886 the first modern production car was patented by Karl Benz.(1886-1893)
Today it known Known as Mercedes Benz
It was having 1 litter gasoline engine and produces an output of 2/3rd HP



The widely made available car was in masses in 20th century by Ford Company that was Model T

CLASSIFICATION OF VEHICLE

Vehicles are classified on the basis different criteria;

1. On the basis purpose

i) Passenger Vehicle
ii) Goods vehicle
iii)Military vehicle
iv)Sports vehicle
v) Industrial vehicle
vi)Other special purpose vehicle

2. On the basis of fuel

- i) Petrol
- ii) Diesel
- iii) LPG
- iv) CNG
- v) Electric powered
- vi) Hydrogen
- vii) Solar
- viii) Steam
- ix) Hybrid

3. On the basis of capacity

i) **Domestic light duty (LMV)**

ii) **Medium Commercial (LCV)**

iii) **Heavy Commercial (HCV)**

4. On the basis of driven wheels

i) **FWD**

ii) **RWD**

iii) **AWD**

iv) **4WD**

5. Based on number of wheels
i) Three wheeler
ii) Four wheeler
iii)Six Wheeler
iv)Multi Axle

6. Based on side of drive
i) Left hand drive (US, Russia, Dubai etc)
ii) Right hand drive (India, Australia etc)

7. On the basis of transmission
i) Manual
ii) Automatic
iii) Semi Automatic

VEHICLE TYPES

Saloon Car / Sedan Car

Estate Car / Station Wagon

Hatchback Car

Coupe Car

Convertible Car

Van

Pick Up

SUV

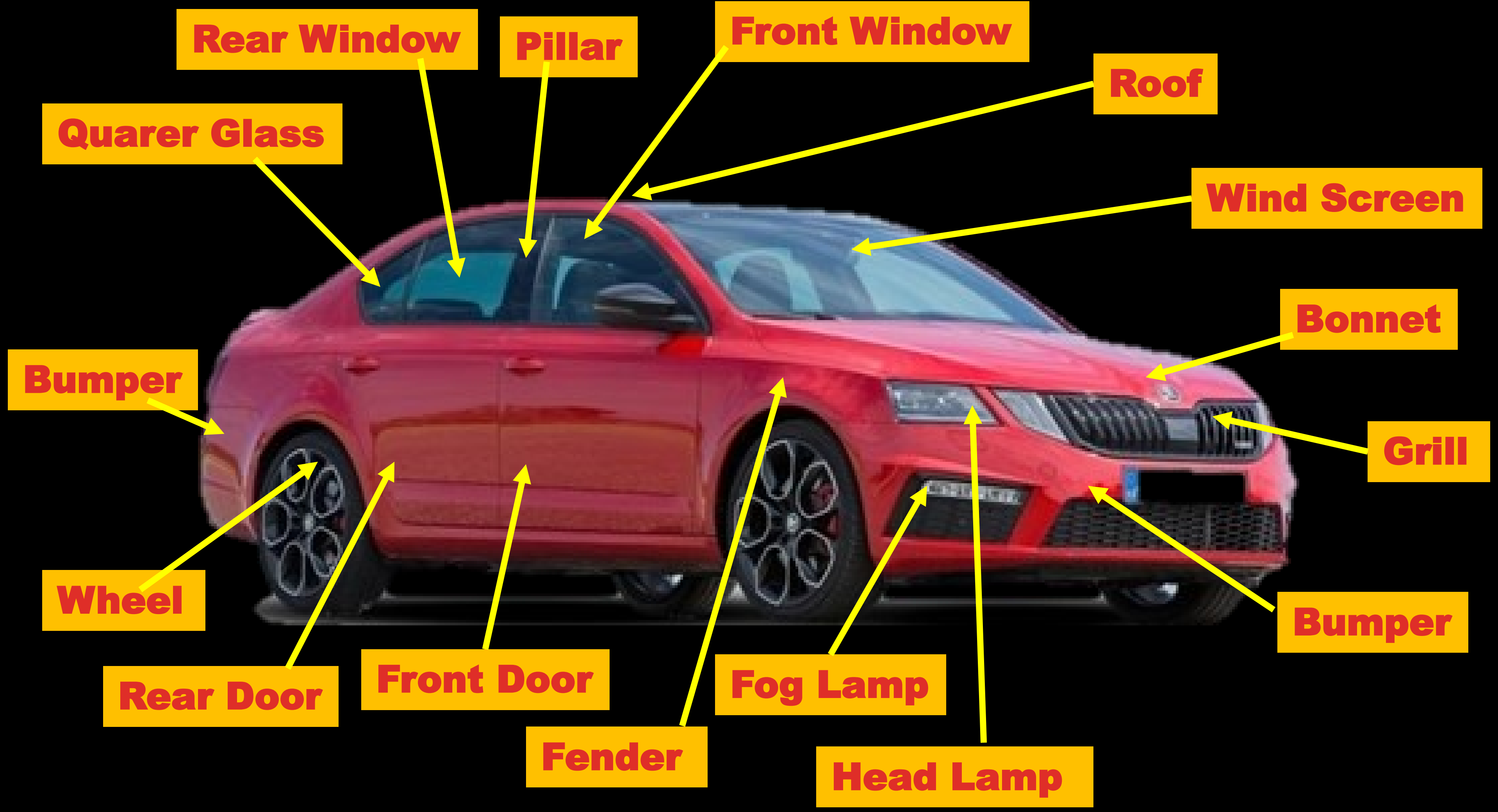
Multi Axle Truck Chassis with TAG Wheel

Half Body Truck

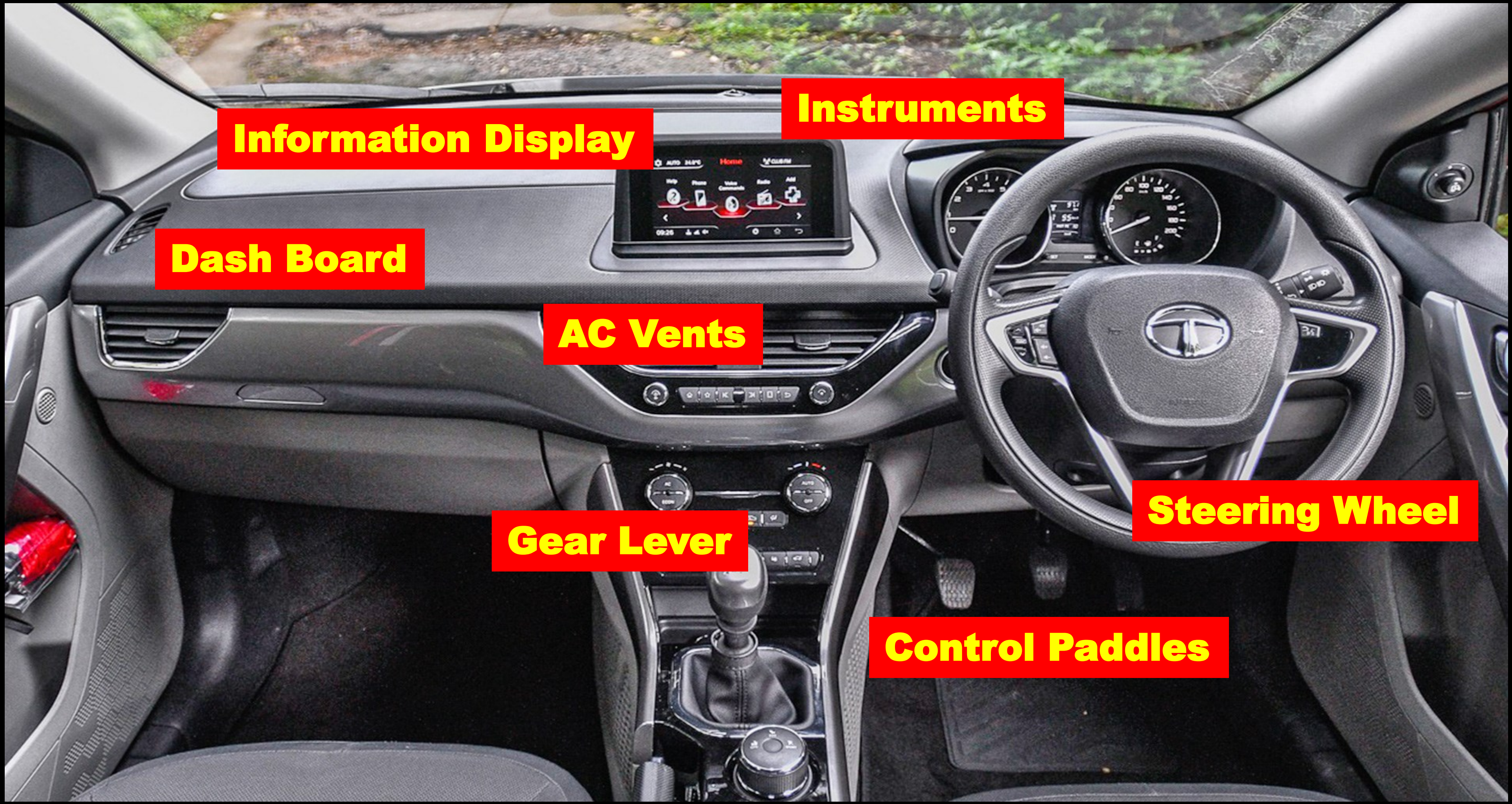
Full Body Truck

Dump Truck

EXTERIOR OF VEHICLE



INTERIOR


Instruments
Information Display
Dash Board
AC Vents
Steering Wheel
Gear Lever
Control Paddles

VEHICLE DESIGN

Vehicle Design Requires Knowledge Various subjects;

1. **Machine Components Design**
2. **Ergonomics**
3. **Human Considerations in design**
4. **Economics**
5. **Manufacturing**
6. **Material Science**
7. **Dynamics and Kinematics**

MAJOR COMPONENTS OF A VEHICLE

The Major Components of Vehicle are

1. **Car Body**
2. **Chassis or Structure**
3. **Power Plant**
4. **Transmission System**
5. **Braking System**
6. **Suspension System**
7. **Steering System**
8. **Wheel**
9. **Air-conditioning System**
10. **Electricals**
11. **Exhaust System**

VEHICLE STRUCTURE

Chassis is a French word was used for structure or frame of the vehicle

Vehicle without body is called as chassis (Structure)

Vehicle frame is the backbone of any vehicle

All the major parts are mounted on the frame or attached with the frame

FUNCTIONS OF THE VEHICLE STRUCTURE

To take loads of passenger or goods carried by the vehicle
To provide support at body, engine, gear box etc.
To take load during braking
To withstand load on acceleration
To bear additional load during cornering
To bear load due to bad road condition
To take load during toeing other vehicle or toed by other vehicle
It bears the load due to collision

LOADS ACTING ON THE FRAME

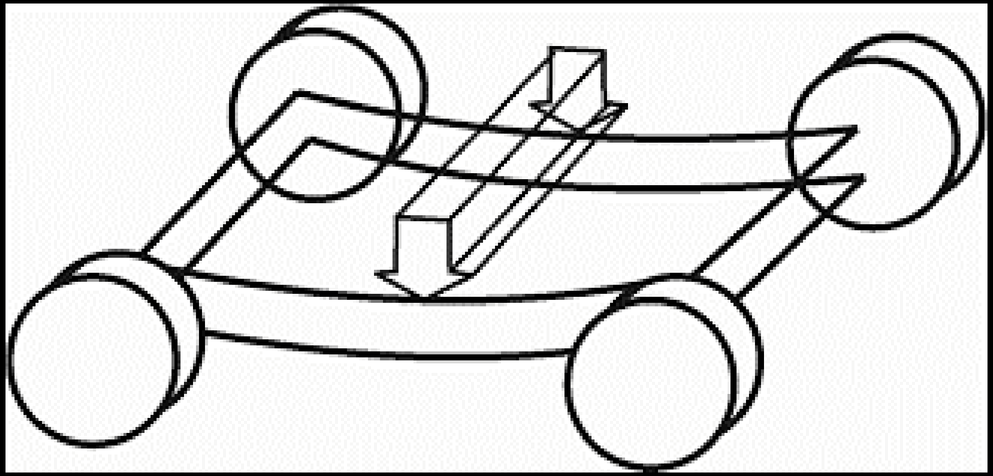
Various loads acts on the vehicle frame as given below;

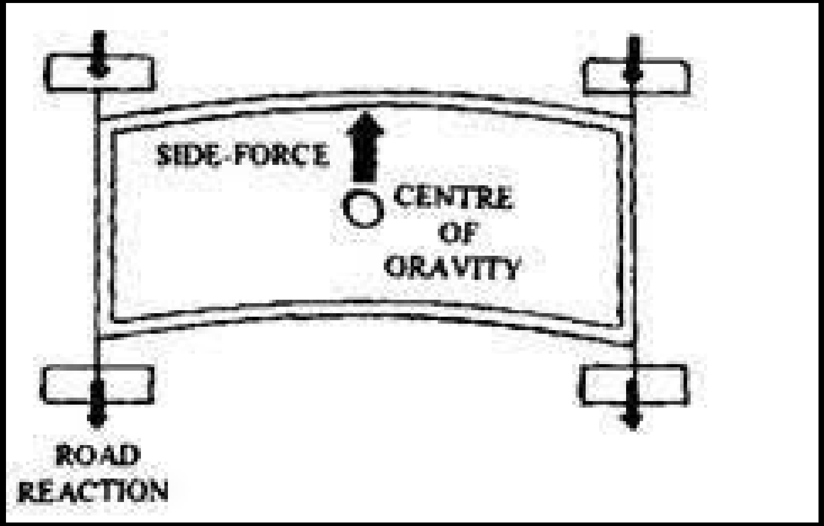
1. Short Duration Load – When vehicle negotiates a curved path
2. Momentary Load – When the any wheel goes into pot or passes over the bump
3. Impact Load – Due to collision of vehicle
4. Inertia Load – While accelerating the vehicle or applying brakes
5. Static Load – Dead load like load of body, engine etc and passenger and luggage load
6. Over load – Any load applied beyond the designed load

EFFECT OF LOADS ON FRAMES

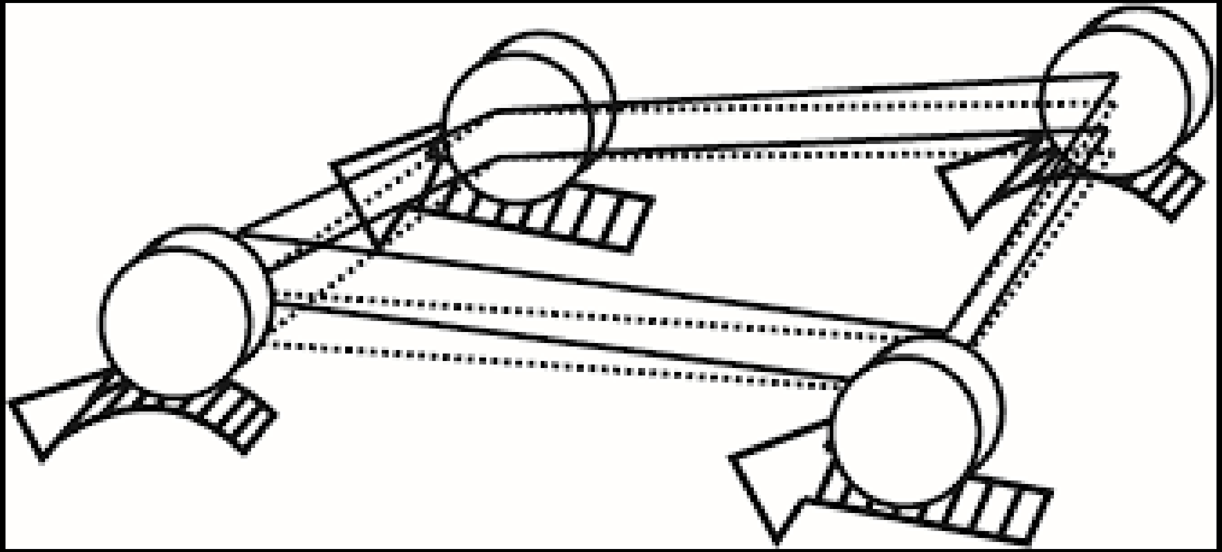
All the frames are subjected to the following two types of loads:

1. Bending of the frame due to various mountings supported by the frame and loads due to driving conditions





2. Torsion of the frame due to position of all the wheels at different levels due to uneven road surface.



TYPES OF VEHICLE FRAME

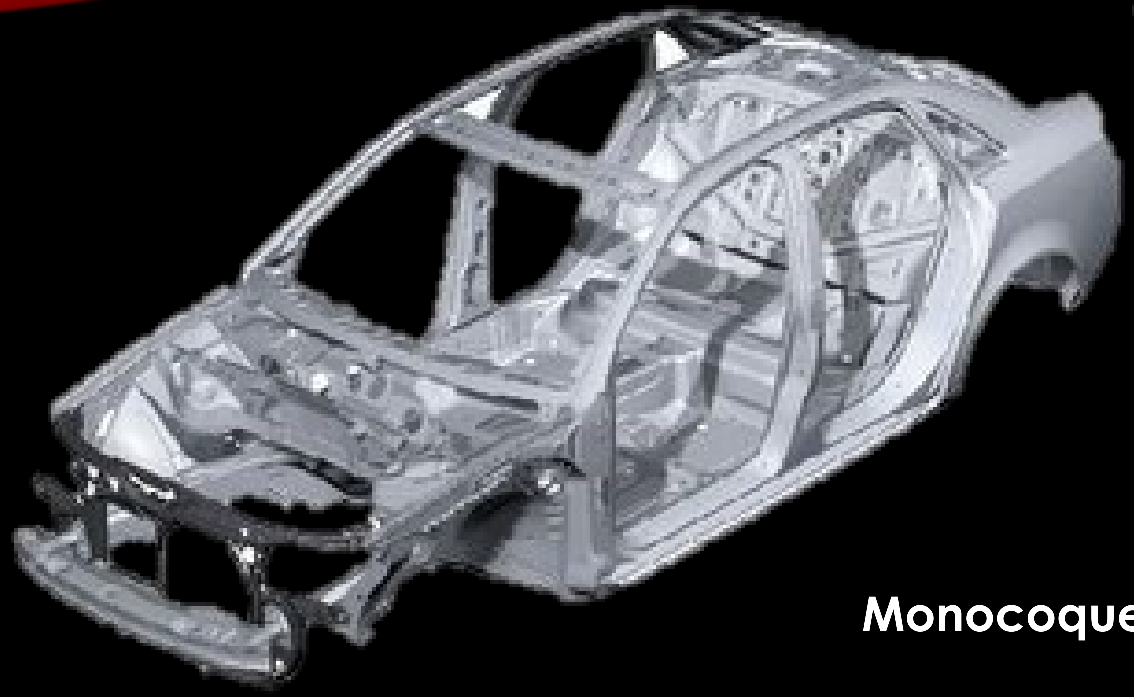
Following are the different types of frames used;

1. Ladder frame (Body on frame)

2. Integral frame or monocoque frame (body) or unibody frame)

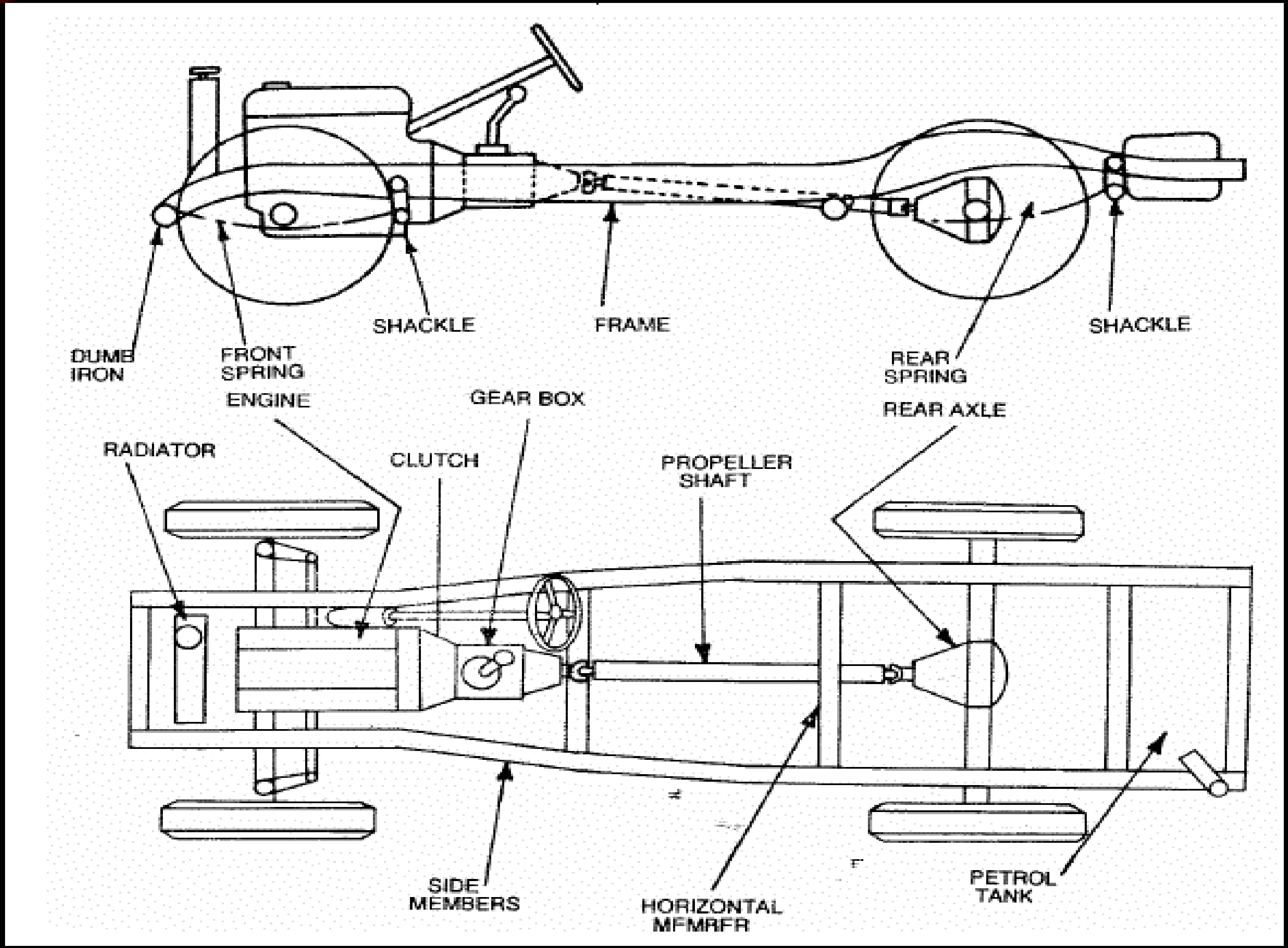
3. Semi Integral Frame

Ladder Frame



Monocoque Frame

LADDER FRAME OR BODY ON FRAME



Ladder frame is shown in figure in the previous slide

Frame has two longitudinal member and 4 or sometimes more cross members makes a ladder types structure. All the members joined together by riveting, bolting or welding.

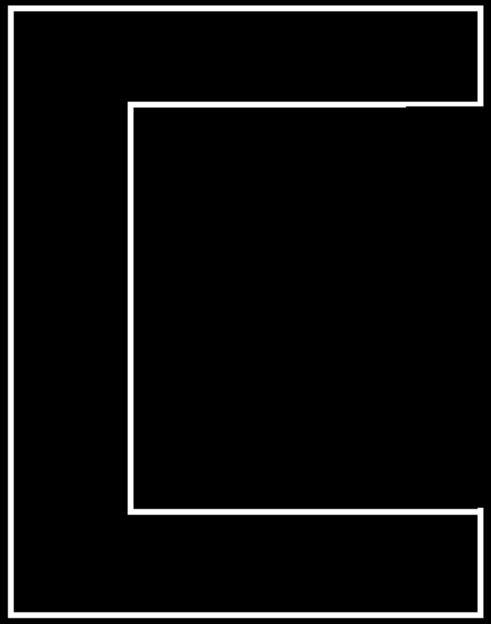
Wheels are connected to the frame through suspension system

Sections of the frame members are:

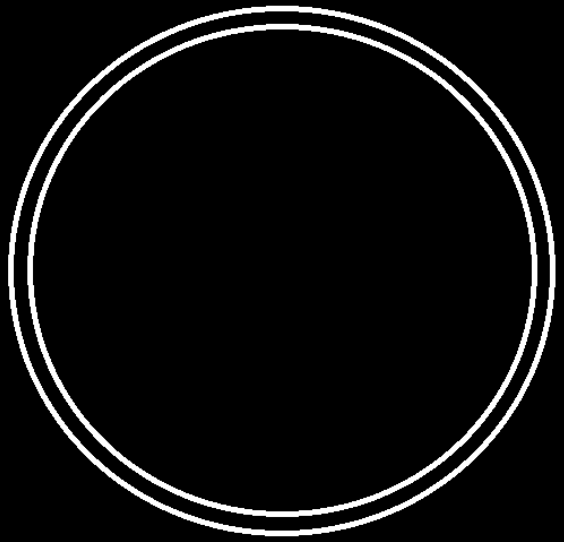
1. Composite section like I or channel section
2. Tubular Section
3. Box section

All the above sections have their own advantages

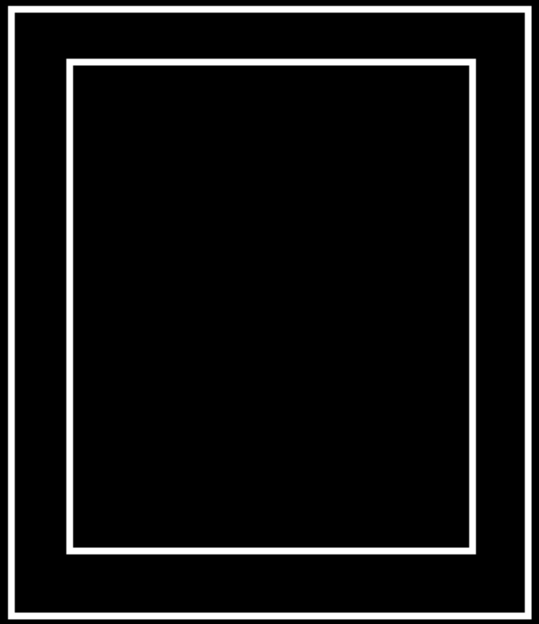
CROSS SECTIONS OF LADDER FRAME MEMBERS



Channel Section
-Less Weight
-Good Bending Resistance
-Longitudinal Members



Tubular Section
-Less Weight
-Good Torsional Resistance
-Cross Members



Box Section
-Less Weight
-Good bending and Torsional Resistance
-Cross and longitudinal Members

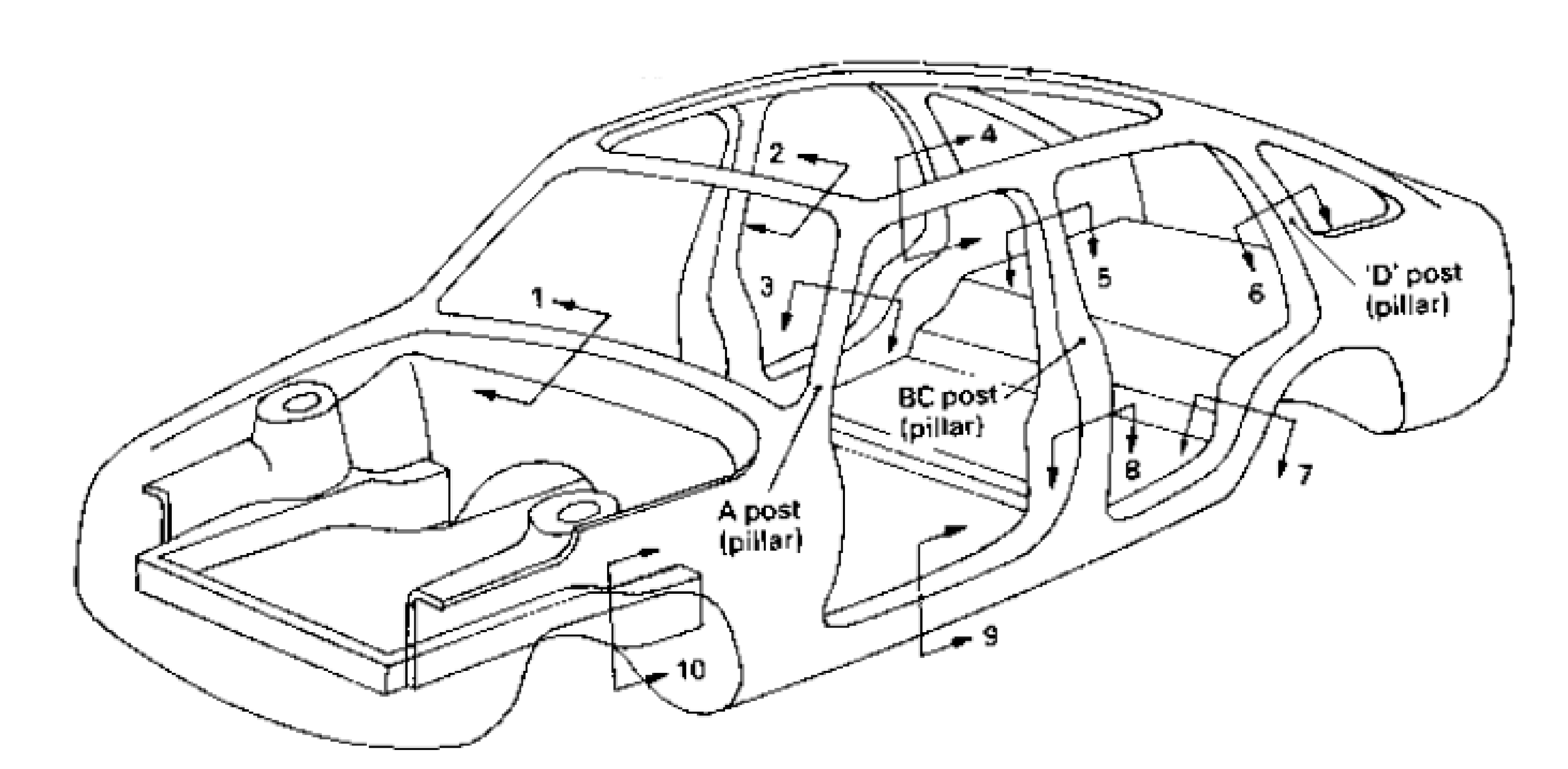
USES OF CONVENTIONAL FRAMES

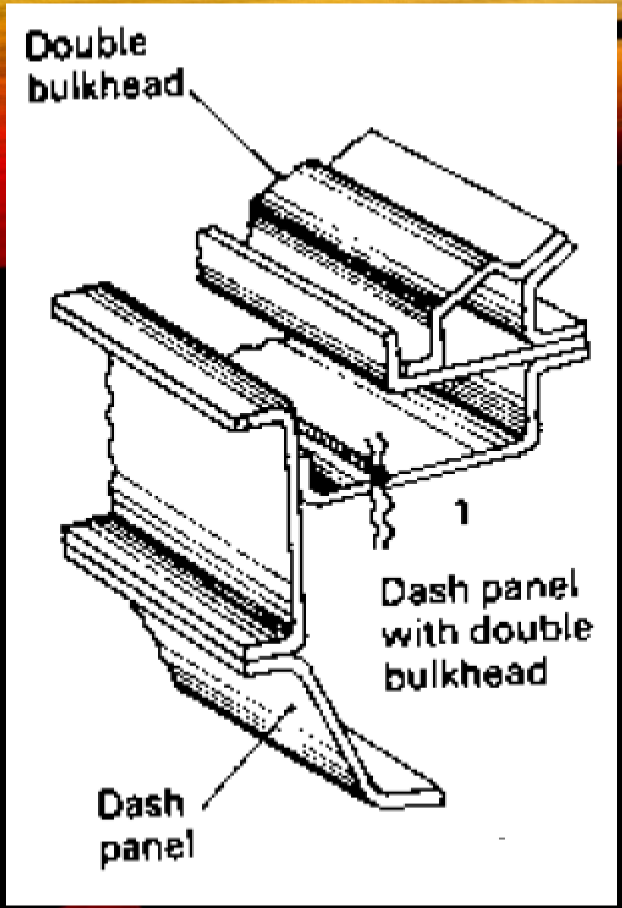
The conventional Frames are used in

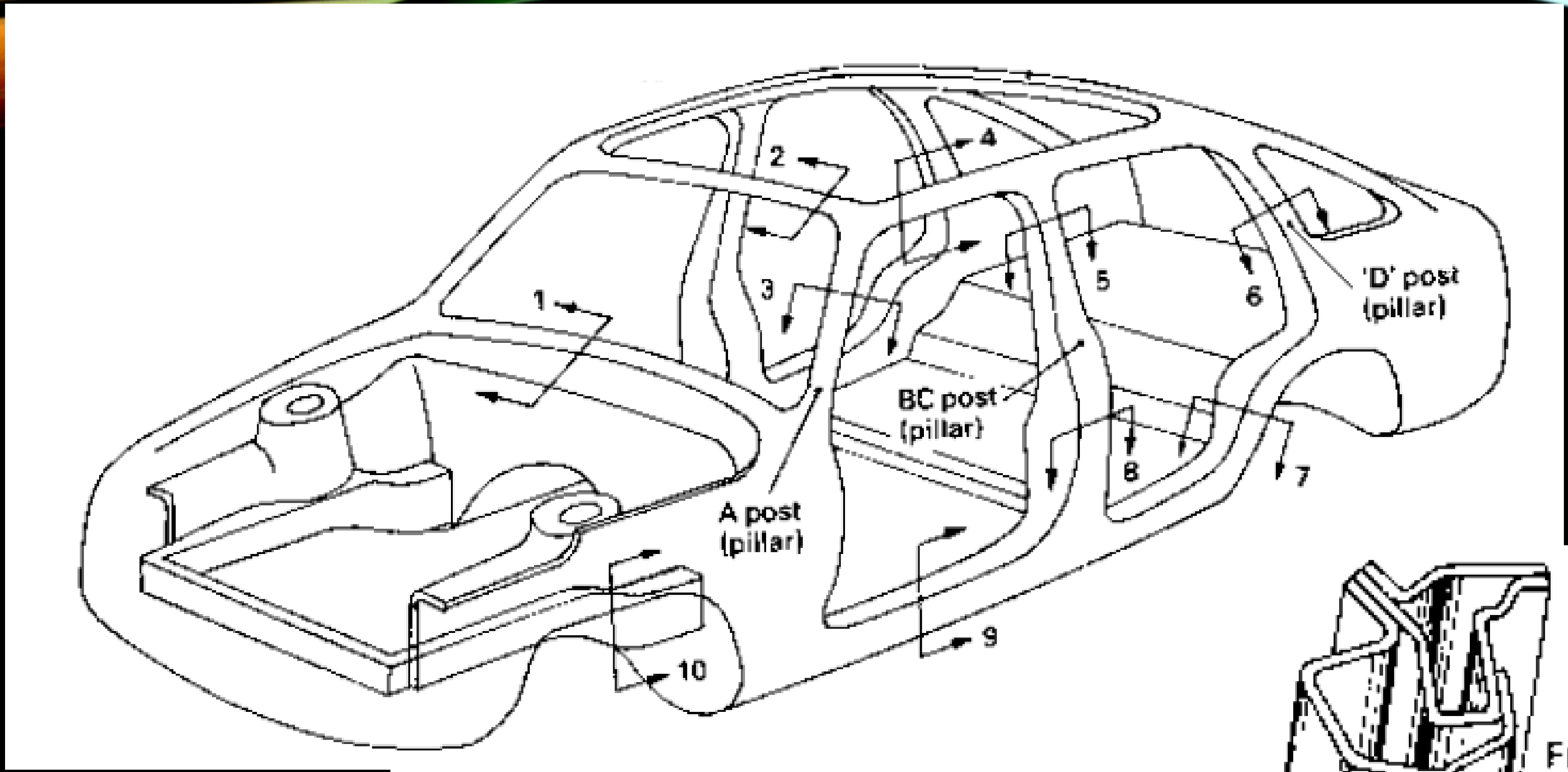
- Trucks
- Buses
- SUVs
- Military jeeps, trucks, buses.

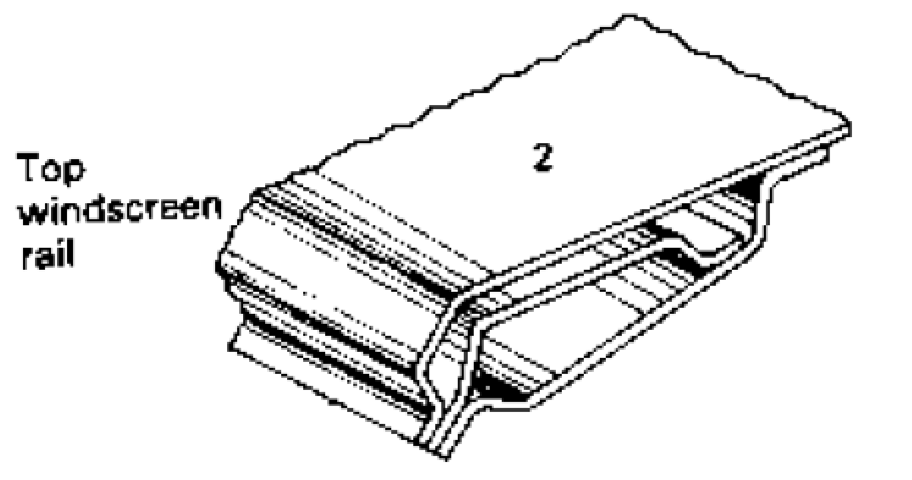
The vehicle body is constructed on the frame by the manufacturer itself or some times it may be fabricated from the local fabricators to meet the customers requirement.

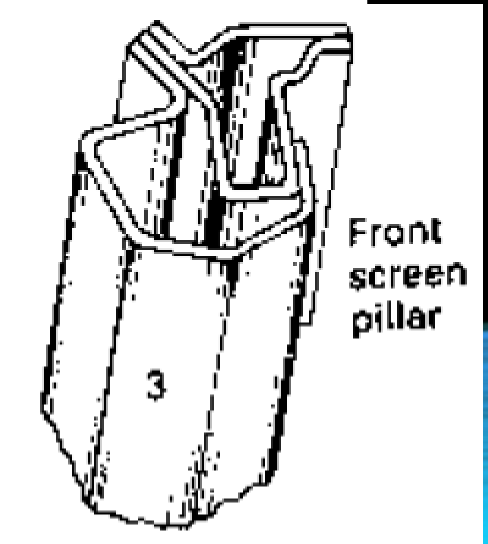
INTEGRAL FRAME

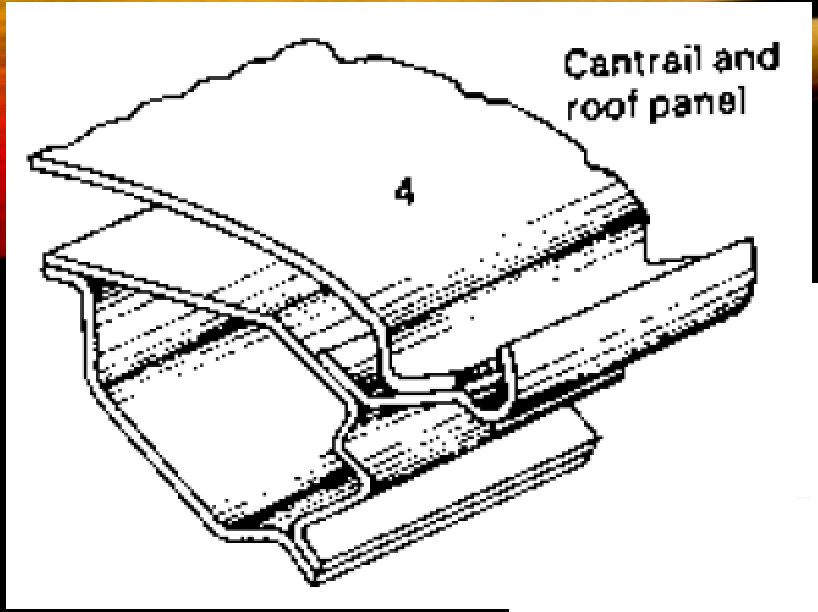


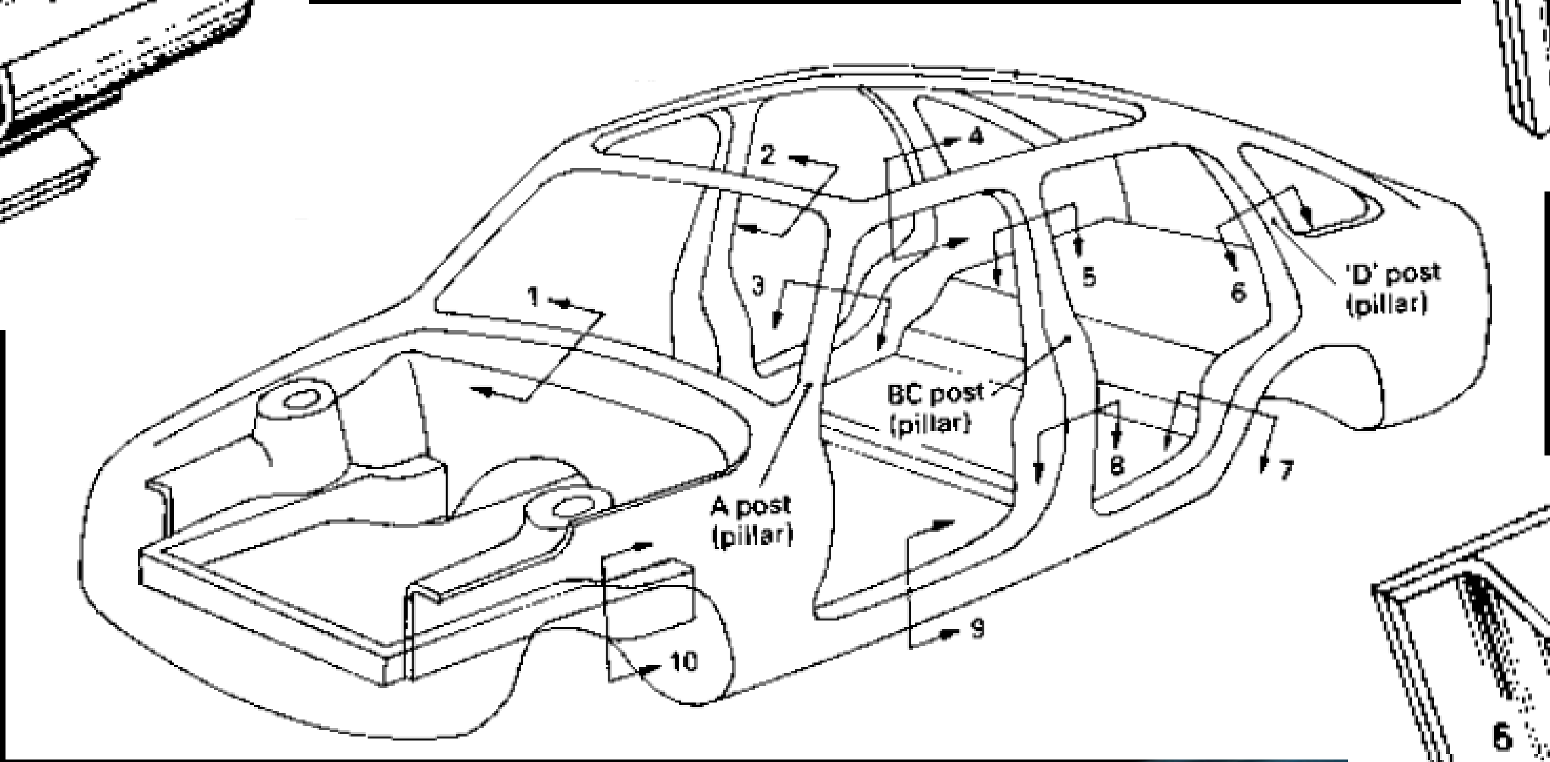

Double bulkhead
1
Dash panel with double bulkhead
Dash panel

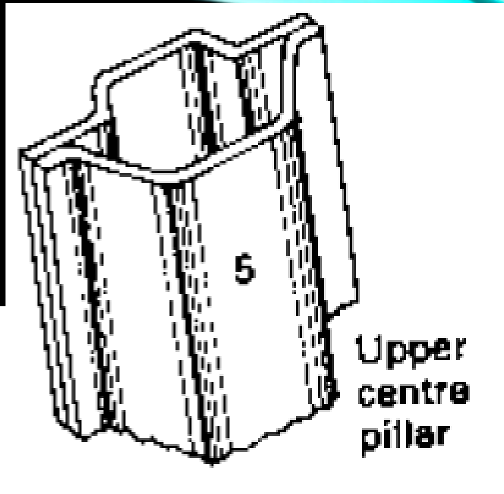

1
2
3
4
5
6
7
8
9
10
'D' post (pillar)
BC post (pillar)
A post (pillar)

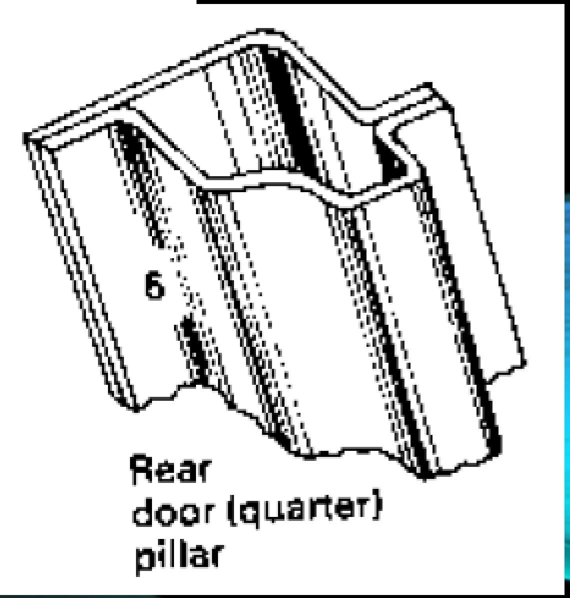

Top windscreen rail
2

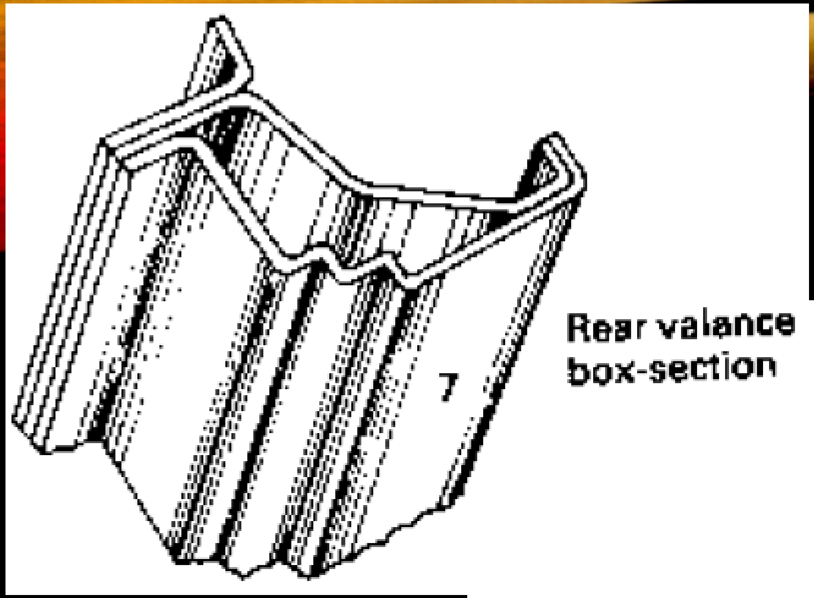

Front screen pillar
3

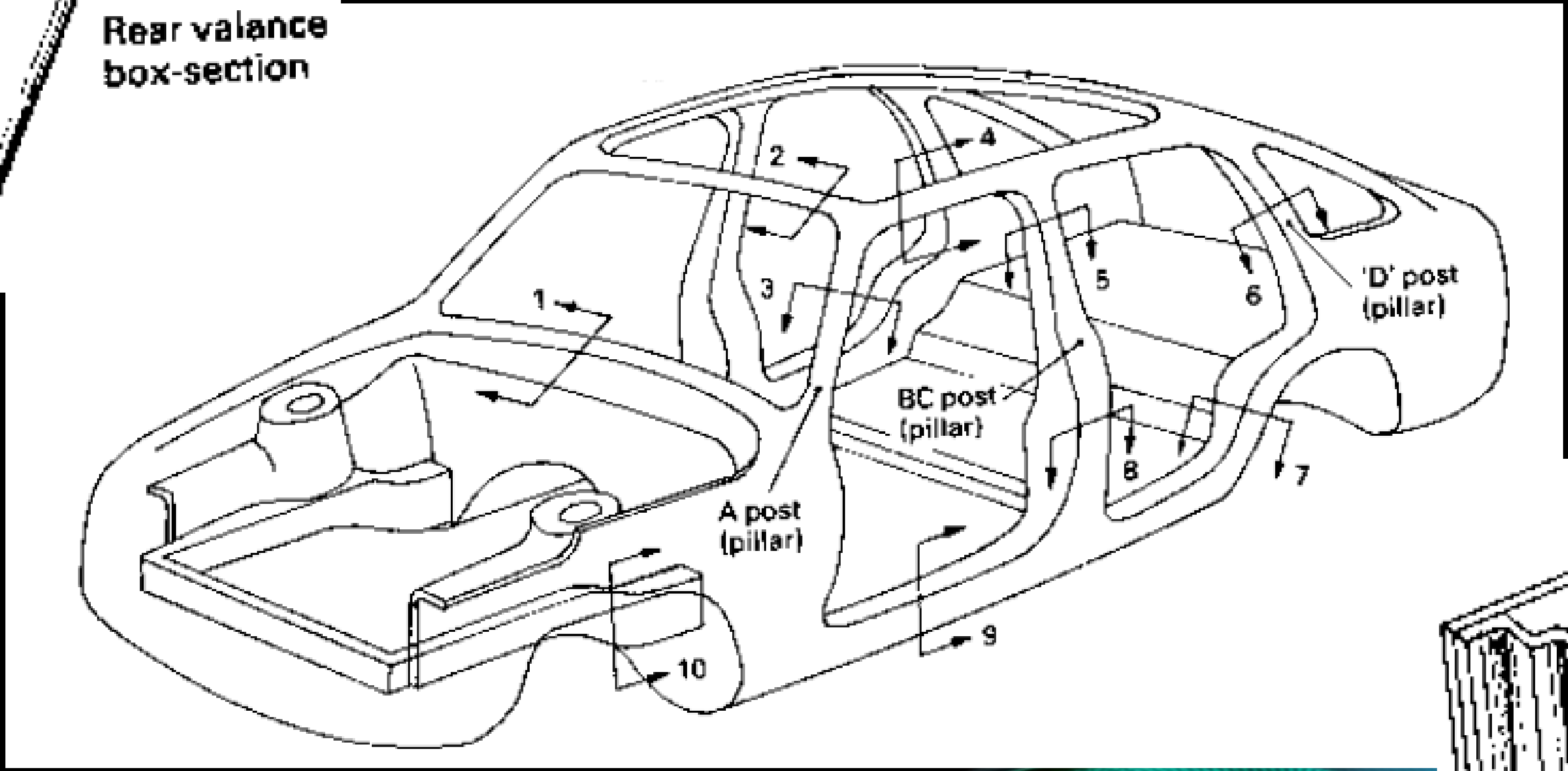

Cantrail and
roof panel
4

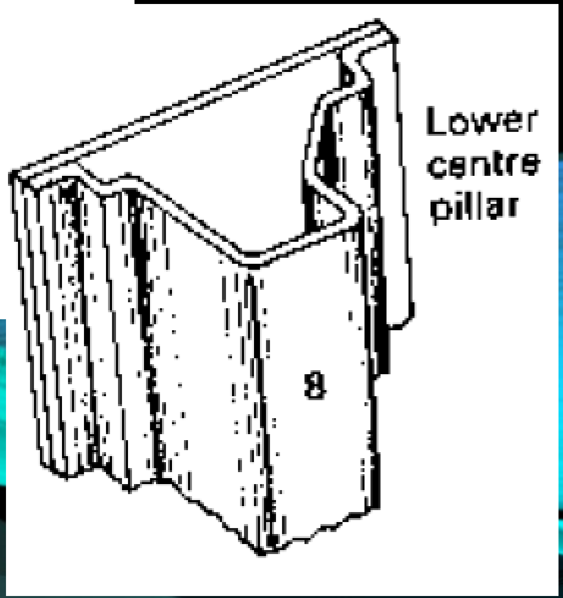

1
2
3
4
5
6
7
8
9
10
A post
(pillar)
BC post
(pillar)
'D' post
(pillar)

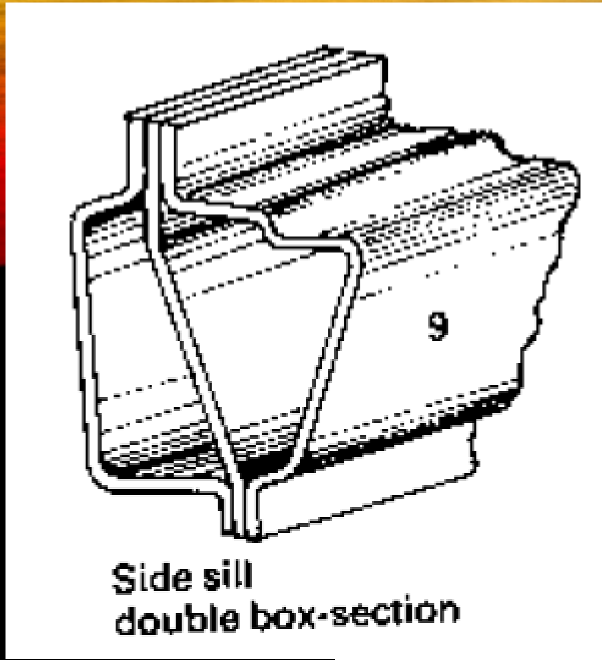

5
Upper
centre
pillar

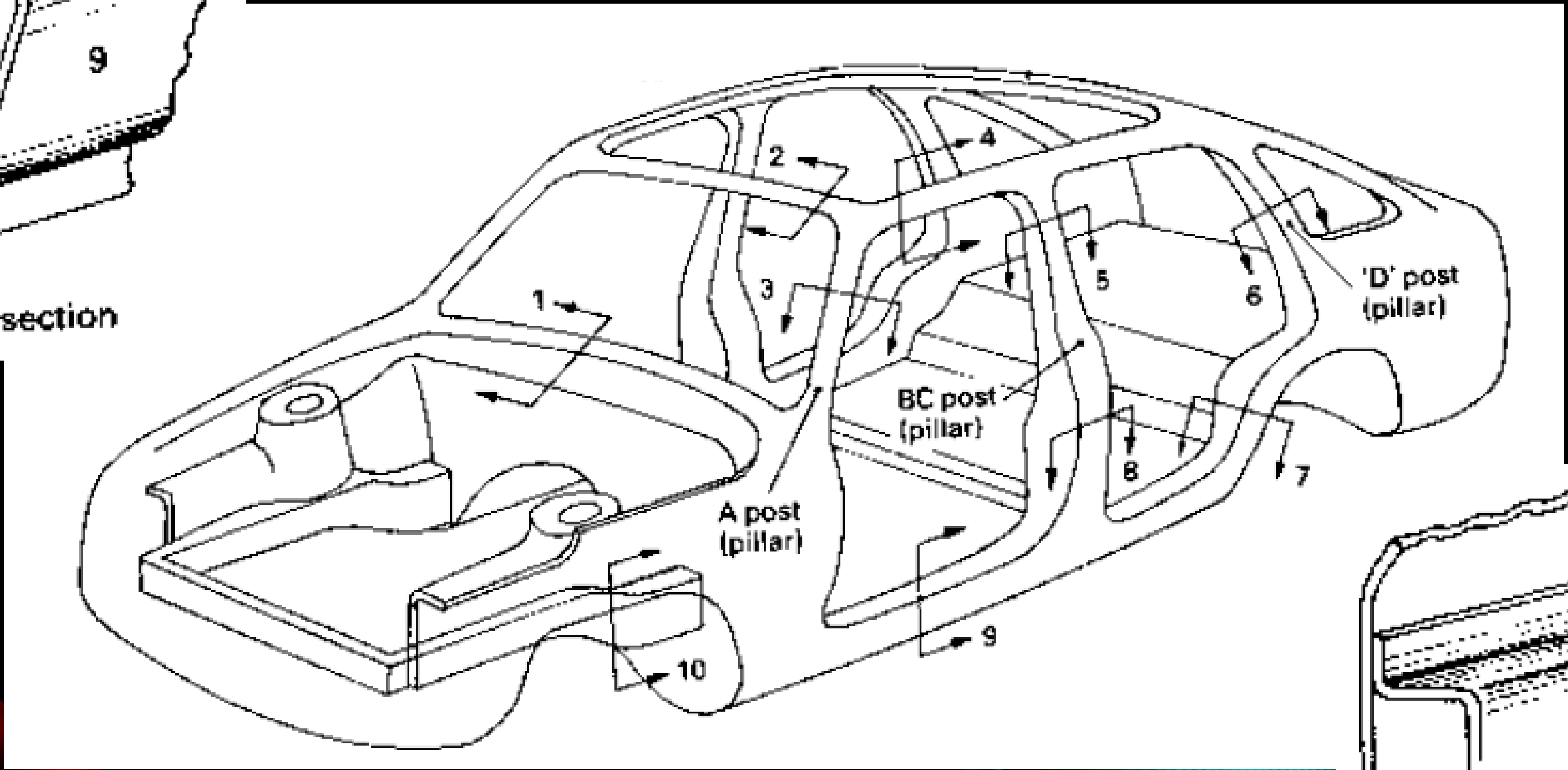

6
Rear
door (quarter)
pillar

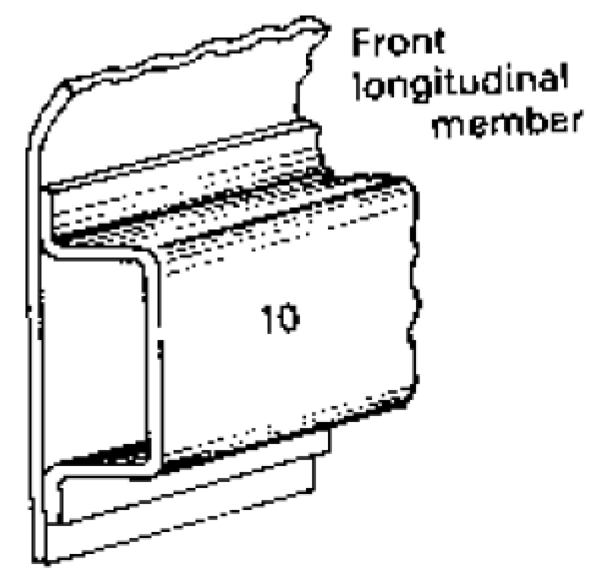

Rear valance
box-section
7

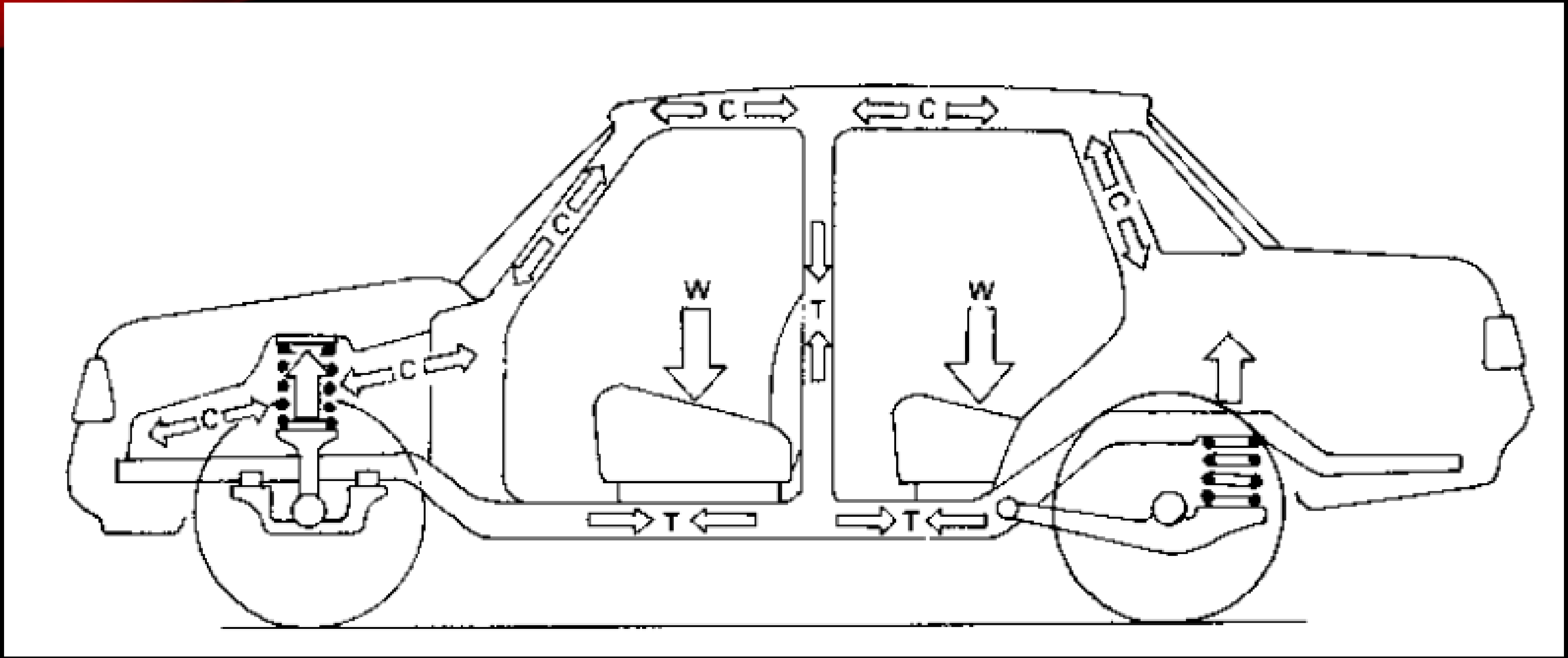

1
2
3
4
5
6
7
8
9
10
A post
(pillar)
BC post
(pillar)
'D' post
(pillar)


Lower
centre
pillar
8


9
Side sill
double box-section

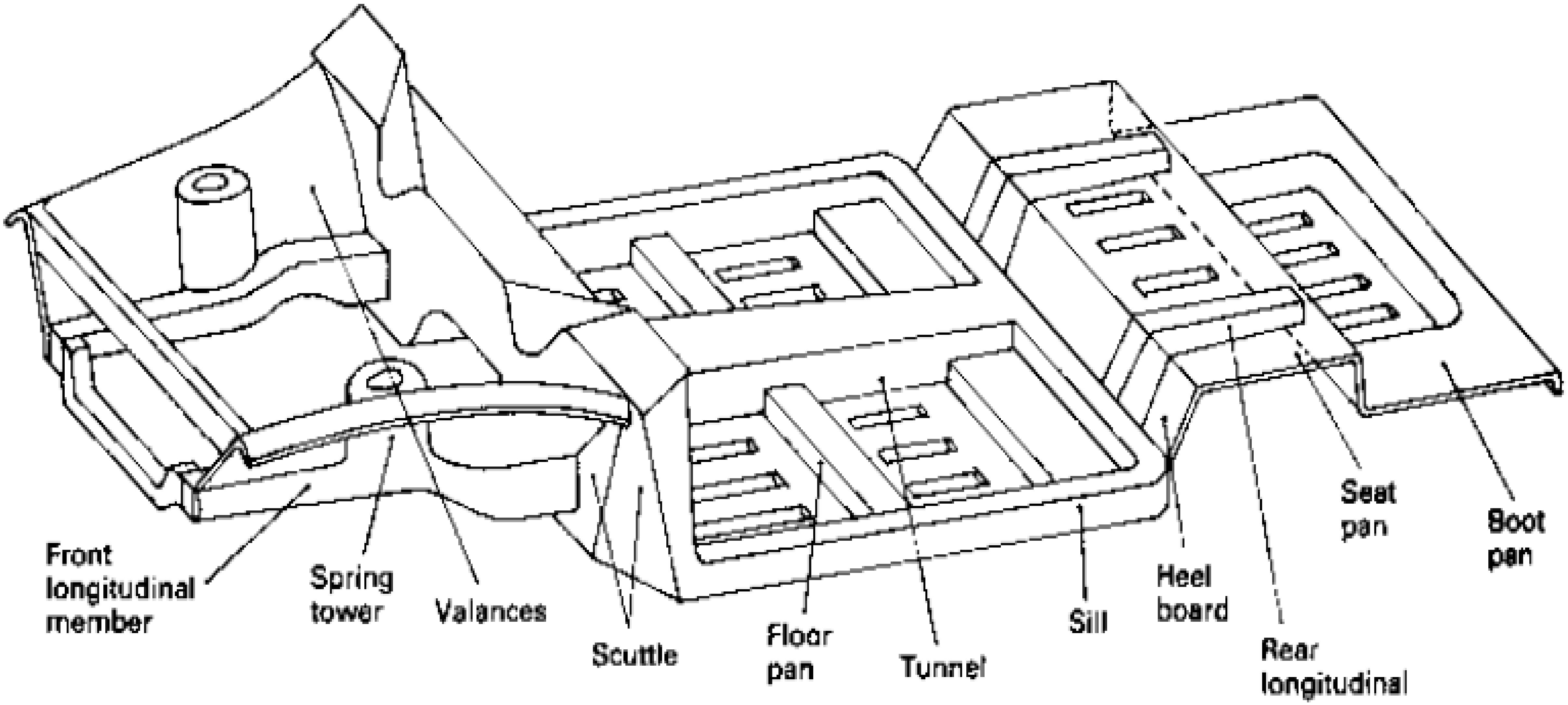

1
2
3
4
5
6
7
8
9
10
A post
(pillar)
BC post
(pillar)
'D' post
(pillar)

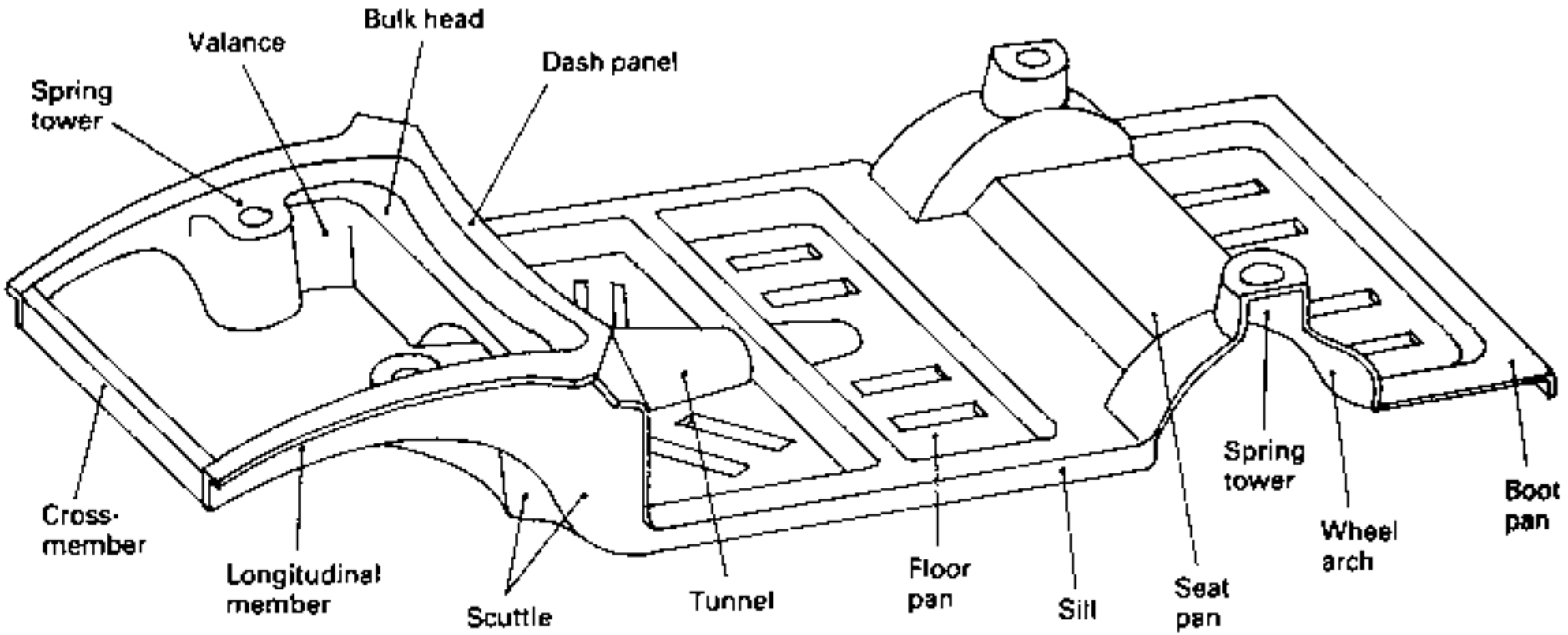

Front
longitudinal
member
10


C
C
C
C
C
C
C
W
W
T
T
T

Parts of Integral Frame

- Window and door pillars
- Windscreen and rear window rails
- Cantrails
- Roof Structure
- Quarter Panel
- Floor seat and boot pan
- Central tunnel
- Sills
- Bulkhead
- Scuttle
- Front longitudinal
- Front Valance
- Rear Valance
- Toe Board
- Heel Board


Front longitudinal member
Spring tower
Valances
Scuttle
Floor pan
Tunnel
Sill
Heel board
Rear longitudinal
Seat pan
Boot pan


Bulk head
Valance
Dash panel
Spring
tower
Cross-
member
Longitudinal
member
Scuttle
Tunnel
Floor
pan
Sill
Seat
pan
Spring
tower
Wheel
arch
Boot
pan

END

Vehicle Safety

Car safety may be broadly divided into two types;

- **Active safety**
- **Passive safety**

Active safety concerns with the car's road holding ability

- **While being driven**
- **Steered**
- **Braked**

Active safety helps to prevent collision to occur or some times reduces severity of collision effect and protects the occupants of the vehicle

Passive safety depends upon

- **Vehicle Body style**
- **Design of vehicle**

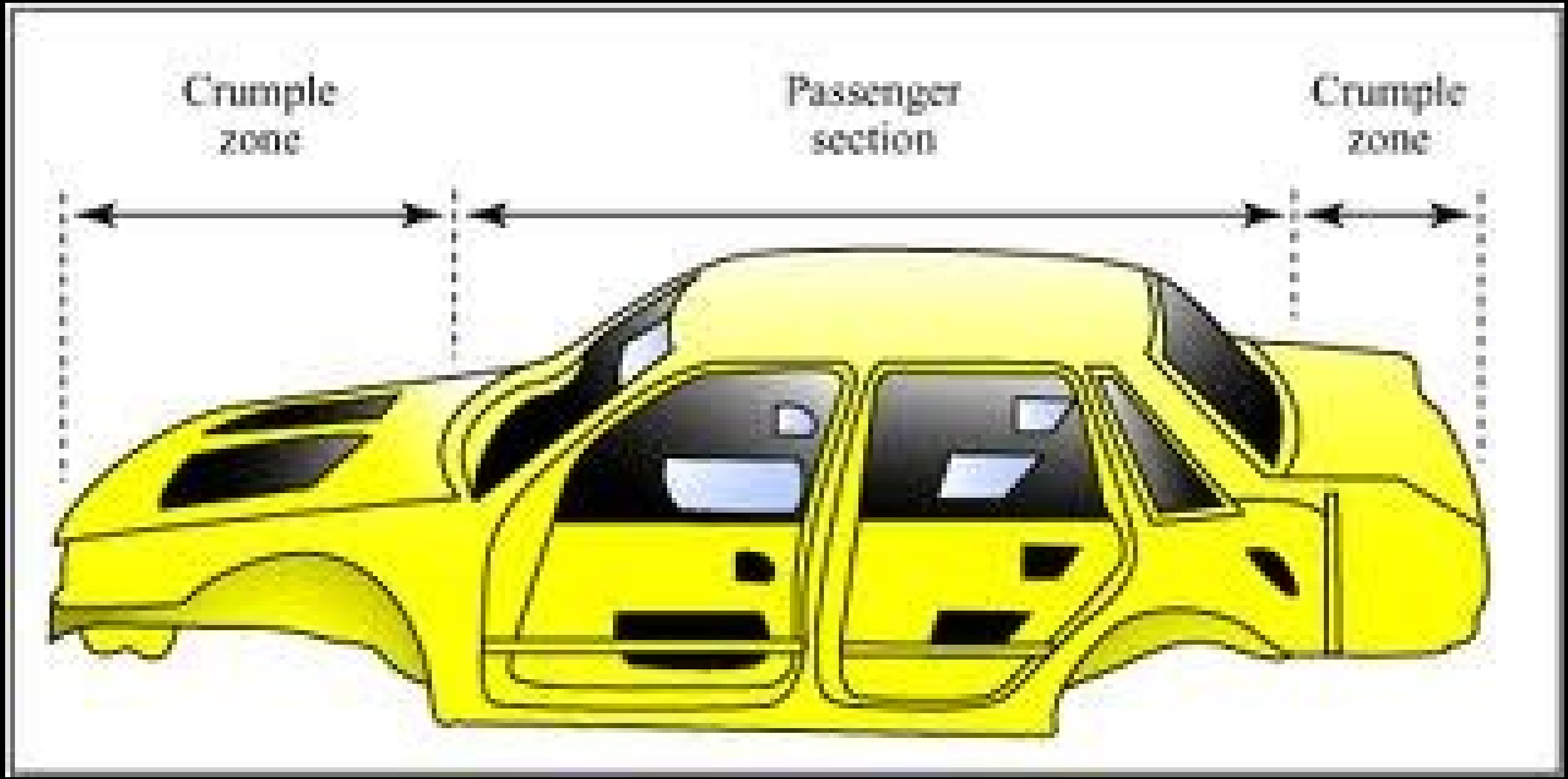
Construction quality of vehicle structure protects the occupants of the car from serious injury in the event of collision.
During collision of vehicle, energy due to collision must be diverted by some means to prevent its passing to passenger.
Passive plays its roll during collision to protect the occupants

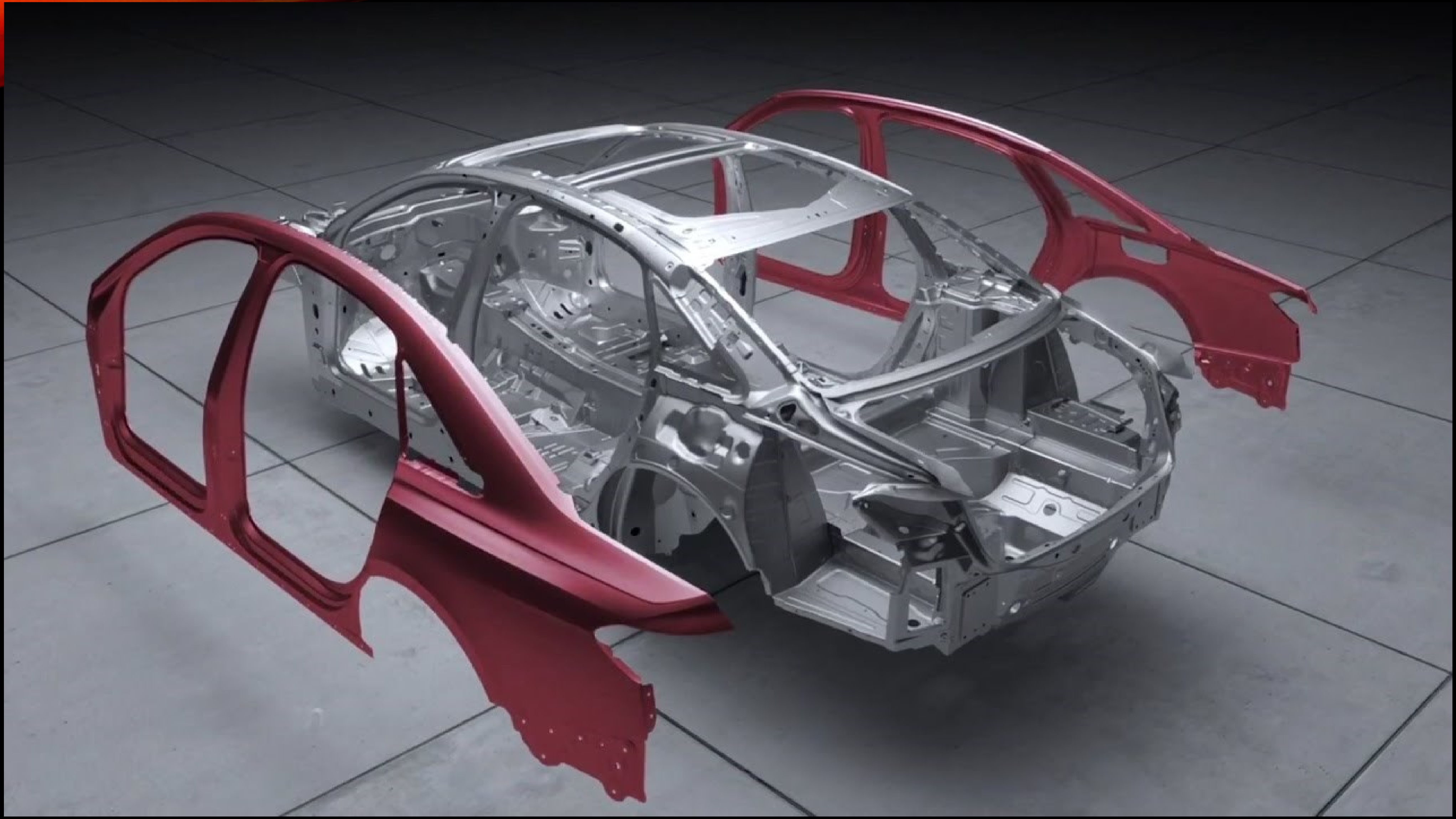
ACTIVE SAFETY

Safety during driving the vehicle, steering and braking ensured by providing the following safety features;

- Antilock braking system (ABS)
- Electronic Brake Force Distribution (EBD)
- Electronic stability control (ESC)
- Lane departure warning system (LDW)
- Adaptive cruse control (ACC)
- Automatic emergency braking system (AEBS) or anti collision system (ACS)
- Rollover mitigation or Rollover stability control (RSC)
- Driver monitoring system (DMS)
- Tyre pressure monitoring system (TPMS)
- Blind spot detection (BSD)
- Night vision system (NVS)
- Head-Up display (HUD)
- Air bags
- Seat belts

Passive safety





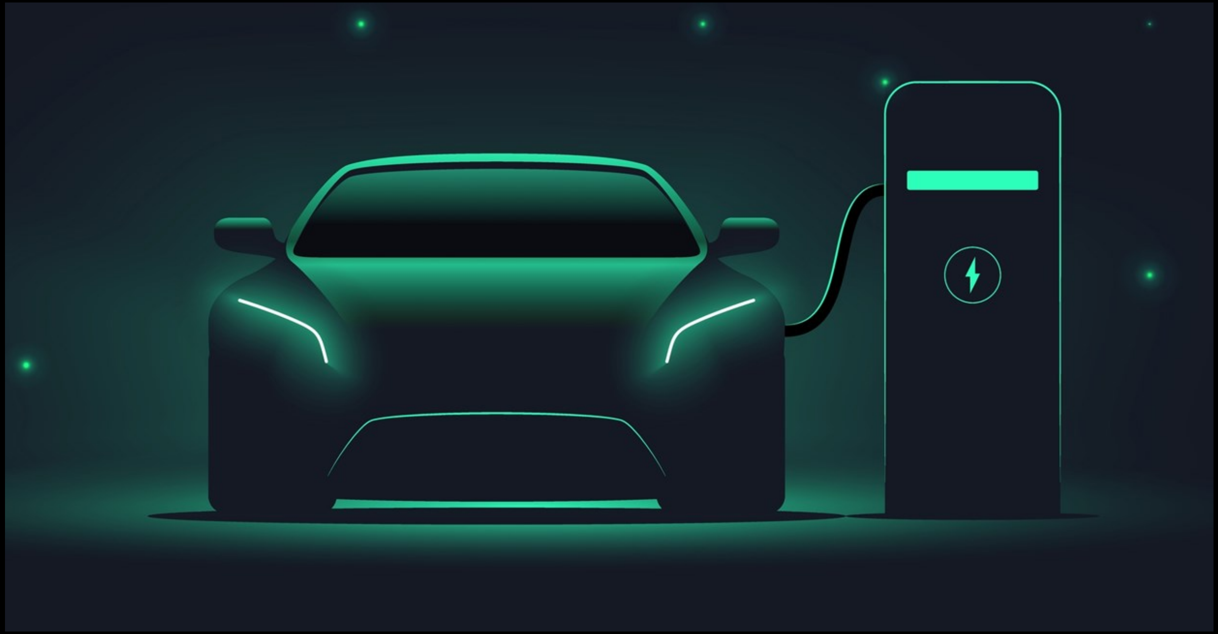
- Nexon Crash Test
- Other Vehicle Test

Electric Vehicle(EV)

Table Contents

Definition

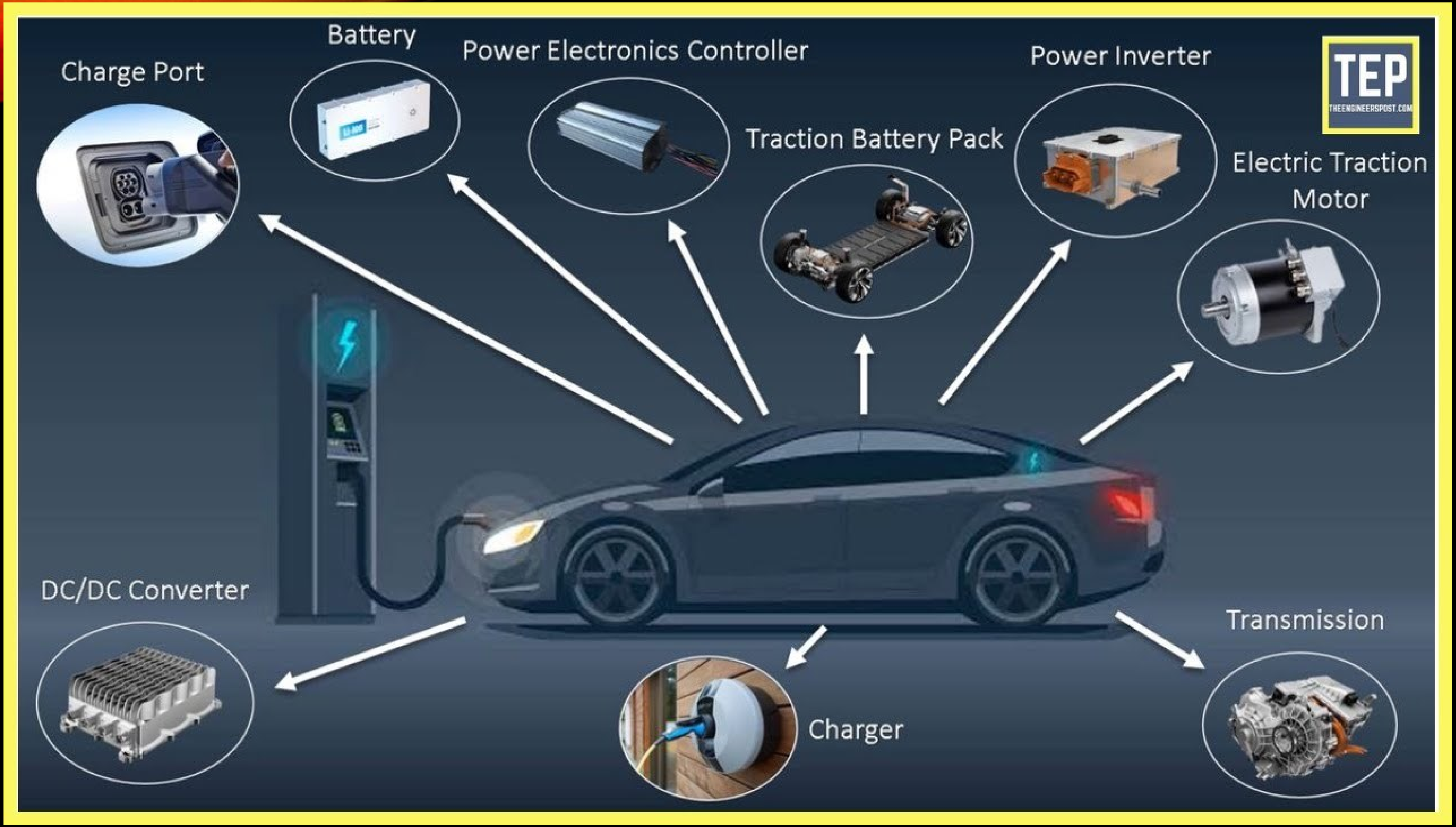
An electric vehicle (EV) is a vehicle that uses one or more electric motors for propulsion and on board battery pack to fulfil the power supply to motor or power is suppiled from overhead wires or from a fuel cell.



Introduction

- It can be powered by a collector system, with electricity from extravehicular sources, or it can be powered autonomously by a battery (sometimes charged by solar panels, or by converting fuel to electricity using fuel cells or a generator).

- EVs include, but are not limited to, road and rail vehicles, surface and underwater vessels, electric aircraft and electric spacecraft.


Charge Port
Battery
Power Electronics Controller
Traction Battery Pack
Power Inverter
TEP
THEENGINEERSPOST.COM
Electric Traction Motor
DC/DC Converter
Charger
Transmission

TYPE OF ELECTRIC MOTOR

AC Motor
- Induction Motor
- PMSM

DC Motor
- Brushed Motor
- Brushless Motor

AC MOTOR

- AC motors play a vital role in many modern electric vehicles, offering high efficiency and robust performance, especially at higher speeds.
- These motors, which operate using alternating current, come in various types, each with distinct characteristics suited to different applications within the EV industry.

Induction Motors: AC motors operate on the principle of electromagnetic induction. They are known for their robustness and ability to deliver high power at varying speeds, making them a popular choice for many EVs. Induction motors offer advantages such as simplicity, durability, and relatively low cost.

1. Power to weight ratio is high.
2. Suitable for operation at high voltage.
3. Suitable for high speed operation.
4. Power converter is simple and economical.
5. Speed control is easy and of low cost.
6. Reliability is good.

Permanent Magnet Synchronous Motors (PMSM): PMSMs use permanent magnets embedded in the rotor to create a constant magnetic field, resulting in high efficiency and precise control. These motors are commonly used in high-performance EVs due to their superior performance and efficiency at high speeds.

Advantages of PMS Motor

- High power density
- Much less noise
- Dynamic performance in both high and low-speed operation
- High-efficiency at high speeds
- Efficient dissipation of heat
- Low rotor inertia makes it easy to control
- No torque ripple when the motor is commutated

- High and smooth torque and dynamic performance
- Resistant to wear and tear
- Available in small sizes at different packages
- Easy maintenance and installation than an induction motor
- Capable of maintaining full torque at low speeds
- High reliability

DC MOTOR

DC motors, while more traditional, continue to be an essential component in the EV industry.

Known for their simplicity and ease of control, DC motors are often used in various EV applications, particularly where cost-effectiveness and straightforward integration are priorities.

- **Brushed DC Motors**: These motors use brushes to transmit electrical current to the rotating part of the motor. They are simple in design and easy to control, making them cost-effective. However, they require regular maintenance due to brush wear.

Advantage of Brushed DC Motor

- Low cost of construction.
- Simple and inexpensive controller.
- No controller is required for fixed speeds.
- The ideal for the extreme environment due to the lack of electronics.
- Two-wire control.
- Rebuild new brush for extended life.

Disadvantages of Brushed DC Motor

- Periodic maintenance is required.
- Less efficient.
- Electrically noisy.
- Poor heat dissipation due to internal rotor construction.
- Higher rotor inertia limits the dynamic characteristics.
- Speed/torque is moderately flat. At higher speed brush friction increases thus reducing useful torque.
- Lower speed range due to mechanical limitation s on the brushes.
- Brush arcing will generate noise-causing electrical magnetic interference.

Brushless DC Motors (BLDC): BLDC motors eliminate the need for brushes by using electronic commutation. This results in higher efficiency, less maintenance, and longer lifespan. BLDC motors are commonly used in modern EVs due to their reliability and performance.

Advantages

- High Efficiency
- Longer Life Span
- High Power Density
- Precise Speed Control
- Quite Operation

Disadvantages

High Cost, Electronic Control System, Sensor Dependency, Limited High Speed and Torque

BATTERY

Lead-Acid Batteries
- Cheap
- Energy density 30 to 50 Wh/kg

Nickel-Cd Hydride Batteries
- Energy density 50-80 Wh/kg

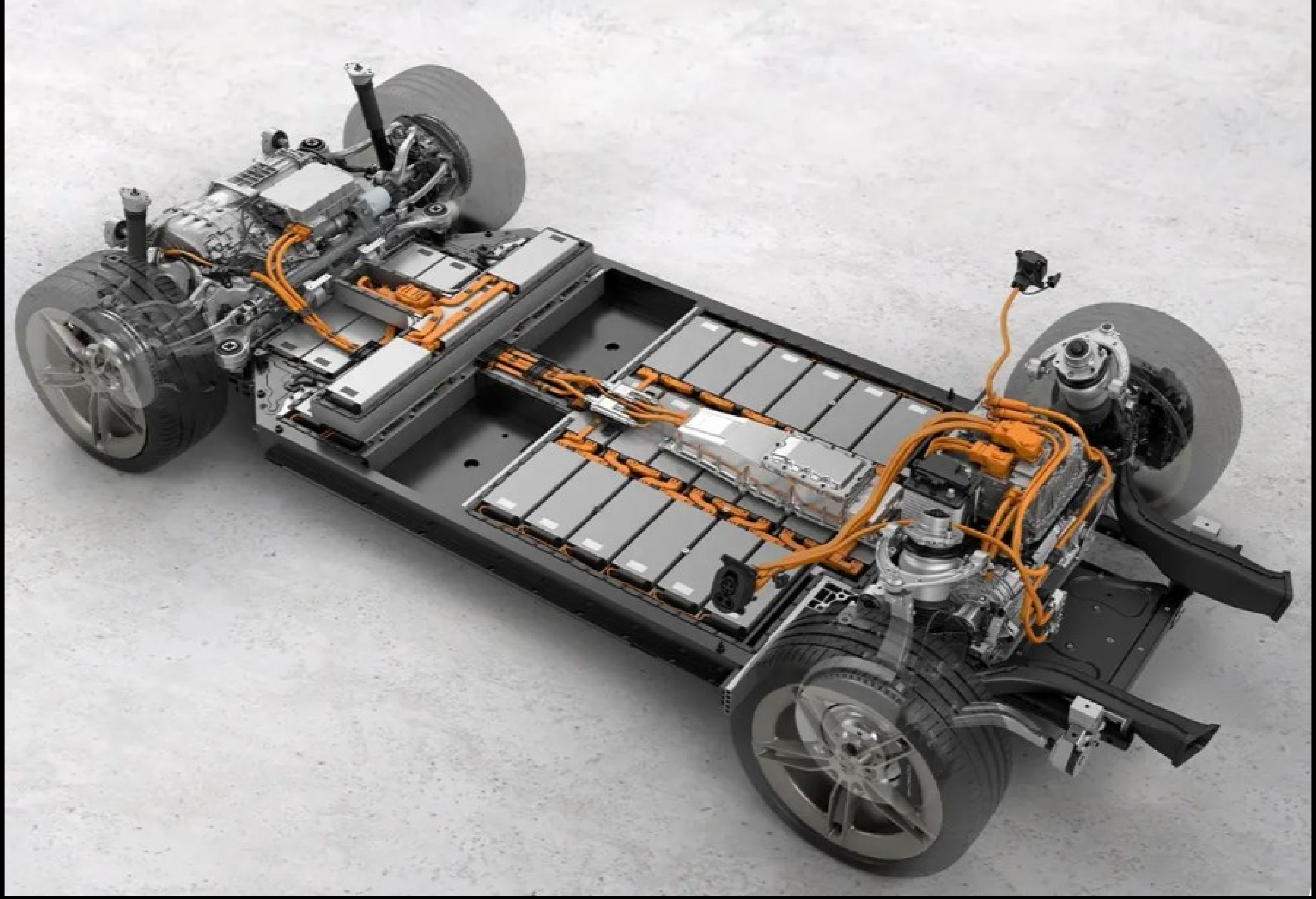
Ultracapacitors
- Energy density 5-15 Wh/kg

Lithium Ion Phosphate
- Energy density 150-300 Wh/kg

Nickel Manganese Cobalt
- Energy density 150-300 Wh/kg

Nickel Cobalt Aluminum
- Energy density 170-200 Wh/kg

Solid-State Batteries
- Energy density 160-250 Wh/kg



Types of Electric Vehicles

Battery Electric Vehicles (BEV)

- Compared to an internal combustion engine, battery powered electric vehicles have fewer moving parts that need maintenance.
- Creates very little noise
- No exhaust, spark plugs, clutch or gears
- Doesn't burn fossil fuels, instead uses rechargeable batteries

Types of Electric Vehicles

- BEVs can be charged at home overnight, providing enough range for average journeys. However, longer journeys or those that require a lot of hill climbs may mean that the fuel cells require charging before you reach your destination, although regenerative braking or driving downhill can help mitigate against this by charging the battery packs.

Types of Electric Vehicles

- The typical charging time for an electric car can range from 30 minutes and up to more than 12 hours. This all depends on the speed of the charging station and the size of the battery.

- In the real world, range is one of the biggest concerns for electric vehicles, but is something that is being addressed by industry.

Types of Electric Vehicles

Plug-in Hybrid Electric Vehicles (PHEV)

- Rather than relying solely on an electric motor, hybrid electric vehicles offer a mixture of battery and petrol (or diesel) power.

- This makes them better for travelling long distances as you can switch to traditional fuels rather than having to find charge points to top up the battery.

Types of Electric Vehicles

- Of course, the same disadvantages that apply to combustion engine vehicles also apply to PHEVs, such as the need for more maintenance, engine noise, emissions and the cost of petrol.

- PHEVs also have smaller battery packs, which means a reduced range.

Benefits of Electric Vehicles

Lower running costs

- The running cost of an electric vehicle is much lower than an equivalent petrol or diesel vehicle.

- Electric vehicles use electricity to charge their batteries instead of using fossil fuels like petrol or diesel.

Top Electric vehicle companies/manufacturers in India

Best electric scooter and top electric bikes in India

TATA MOTORS
TORK MOTORS
Mahindra ELECTRIC
HYUNDAI
ASHOK LEYLAND
MG
Hero
ATHER
LOHIA
OKINAWA Power the Change

Benefits of Electric Vehicles

Low maintenance cost

- Electric vehicles have very low maintenance costs because they don’t have as many moving parts as an internal combustion vehicle.

- The servicing requirements for electric vehicles are lesser than the conventional petrol or diesel vehicles.

Benefits of Electric Vehicles

Tax and financial benefits

- Registration fees and road tax on purchasing electric vehicles are lesser than petrol or diesel vehicles.

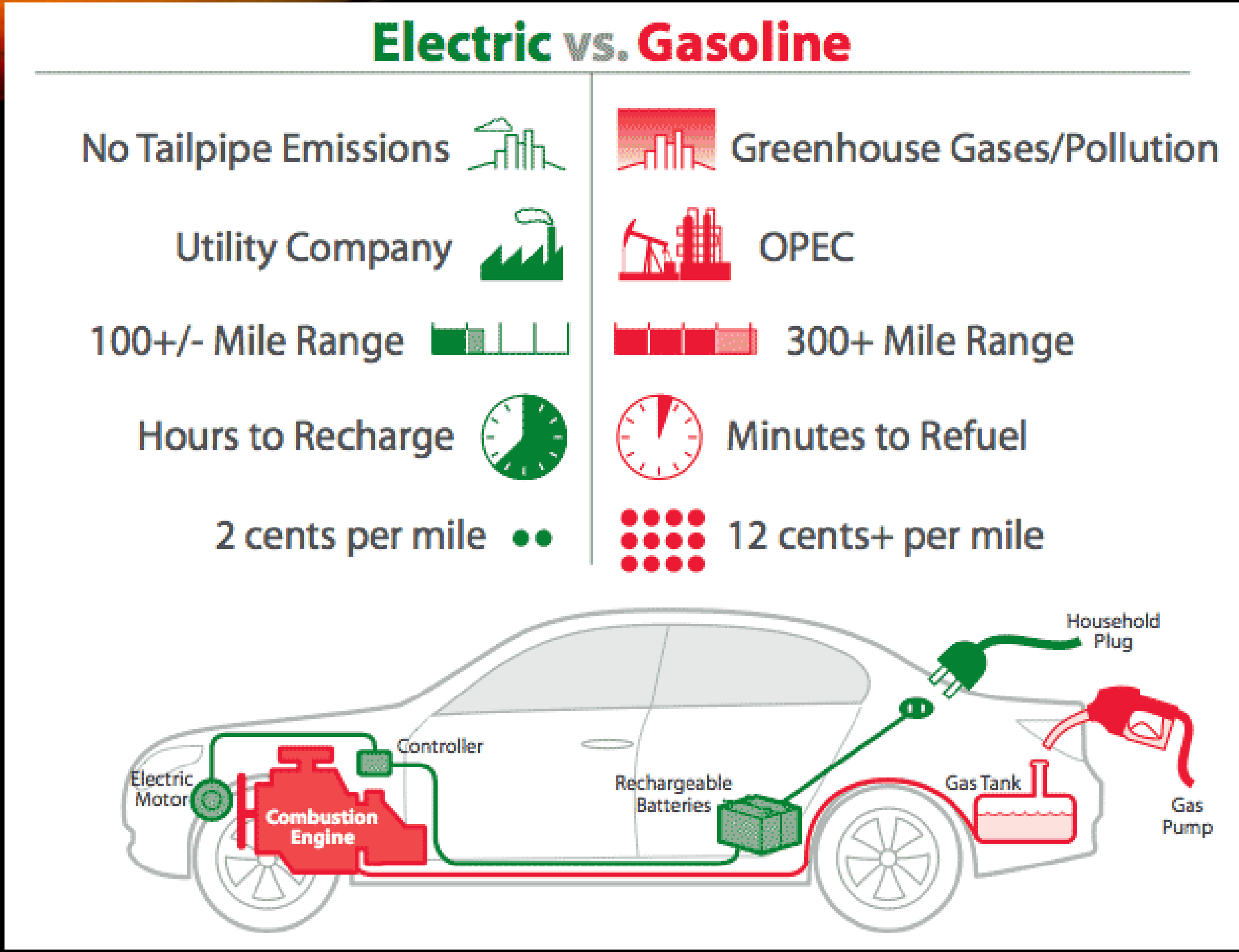
- There are multiple policies and incentives offered by the government depending on which state you are in.

Benefits of Electric Vehicles

No noise pollution

- Electric vehicles have the silent functioning capability as there is no engine under the hood. No engine means no noise.

- The electric motor functions so silently that you need to peek into your instrument panel to check if it is ON.


Electric vs. Gasoline
No Tailpipe Emissions
Greenhouse Gases/Pollution
Utility Company
OPEC
100+/- Mile Range
300+ Mile Range
Hours to Recharge
Minutes to Refuel
2 cents per mile
12 cents+ per mile
Household Plug
Controller
Electric Motor
Combustion Engine
Rechargeable Batteries
Gas Tank
Gas Pump

Conclusion

- ✓ All-electric vehicles, also referred to as battery electric vehicles (BEVs), have an electric motor instead of an internal combustion engine.